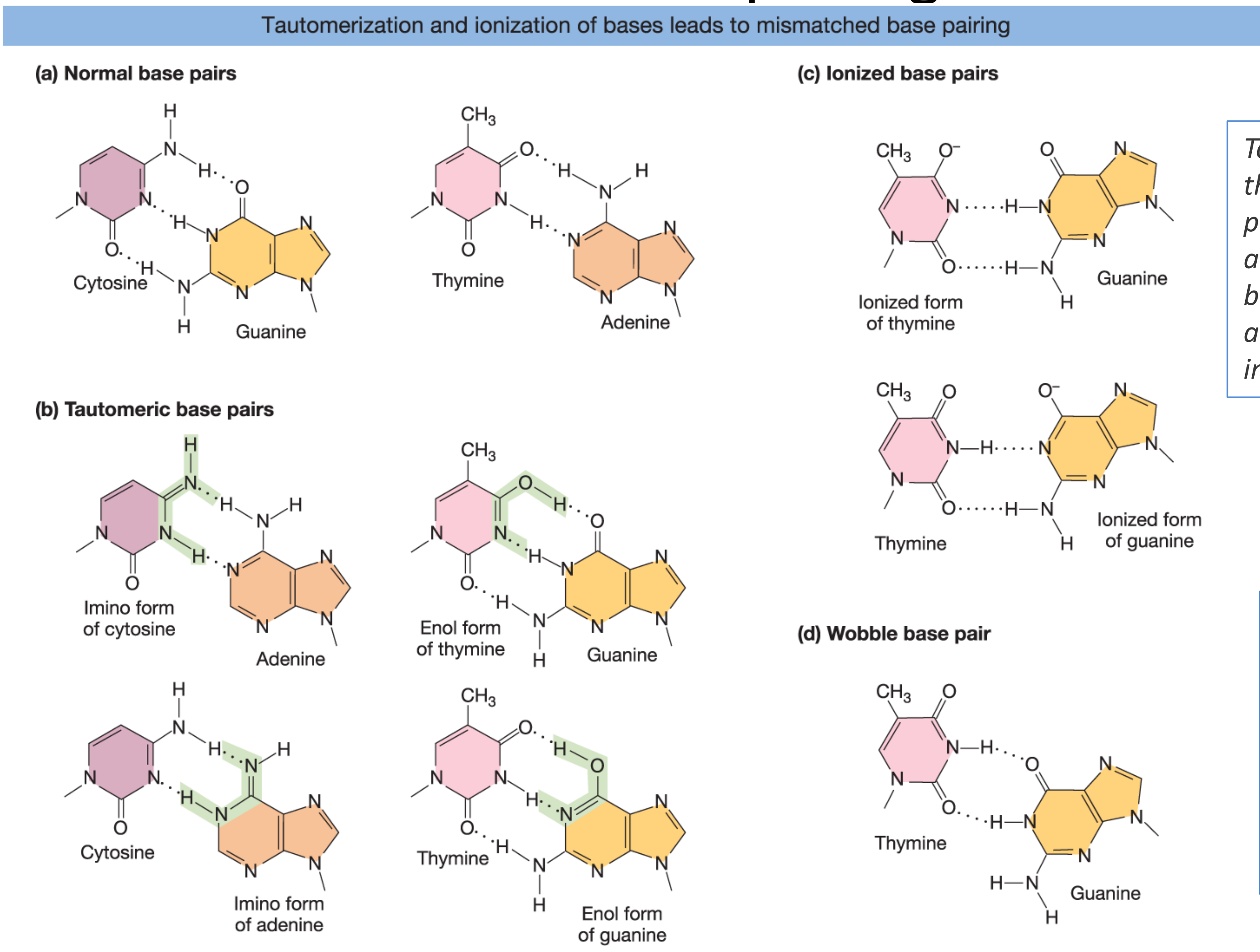
DNA Mutations

Causes and Consequences

DNA replication errors

Base mispairing
Strand slippage

Spontaneous mutations can be caused by errors in base pairing



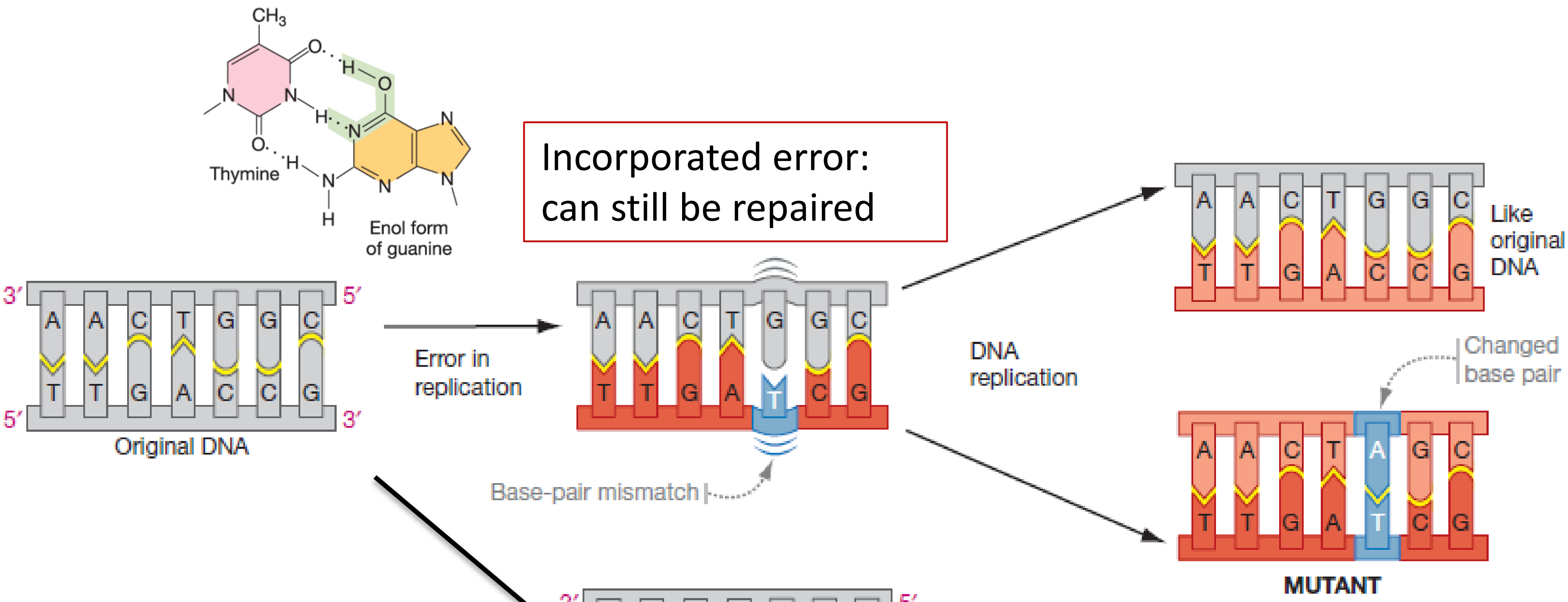
WHEN?

Tautomers: isomers that differ in the positions of their atoms and in the bonds between the atoms. The forms are in equilibrium.

If proofreading does not occur, the mismatches lead to transition mutations

Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020 W. H. Freeman and Company

Spontaneous mutations:
Errors in DNA replication due to errors in base pairing



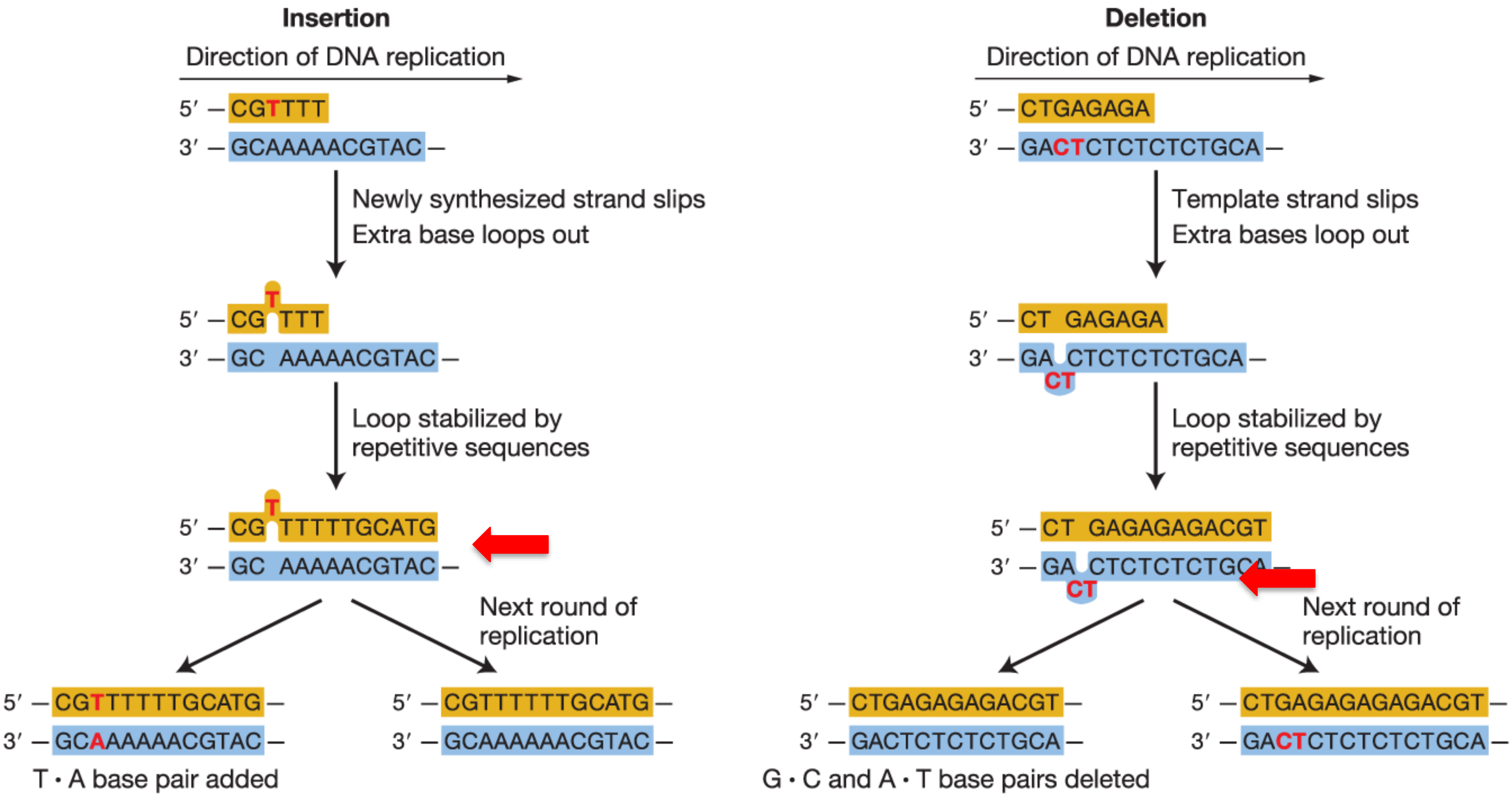
Incorporated error: can still be repaired

Replicated error: permanent

$$\begin{matrix} G \\ C \end{matrix} \rightarrow \begin{matrix} A \\ T \end{matrix}$$

G*C → A*T

Spontaneous mutations: strand slippage during replication



Strand slippage during DNA replication leads to indels.
Whether slippage generates an insertion or deletion depends on which strand loops out.

Spontaneous mutations: strand slippage during replication

Trinucleotide repeat disorders arise from expansion of repeated sequences.

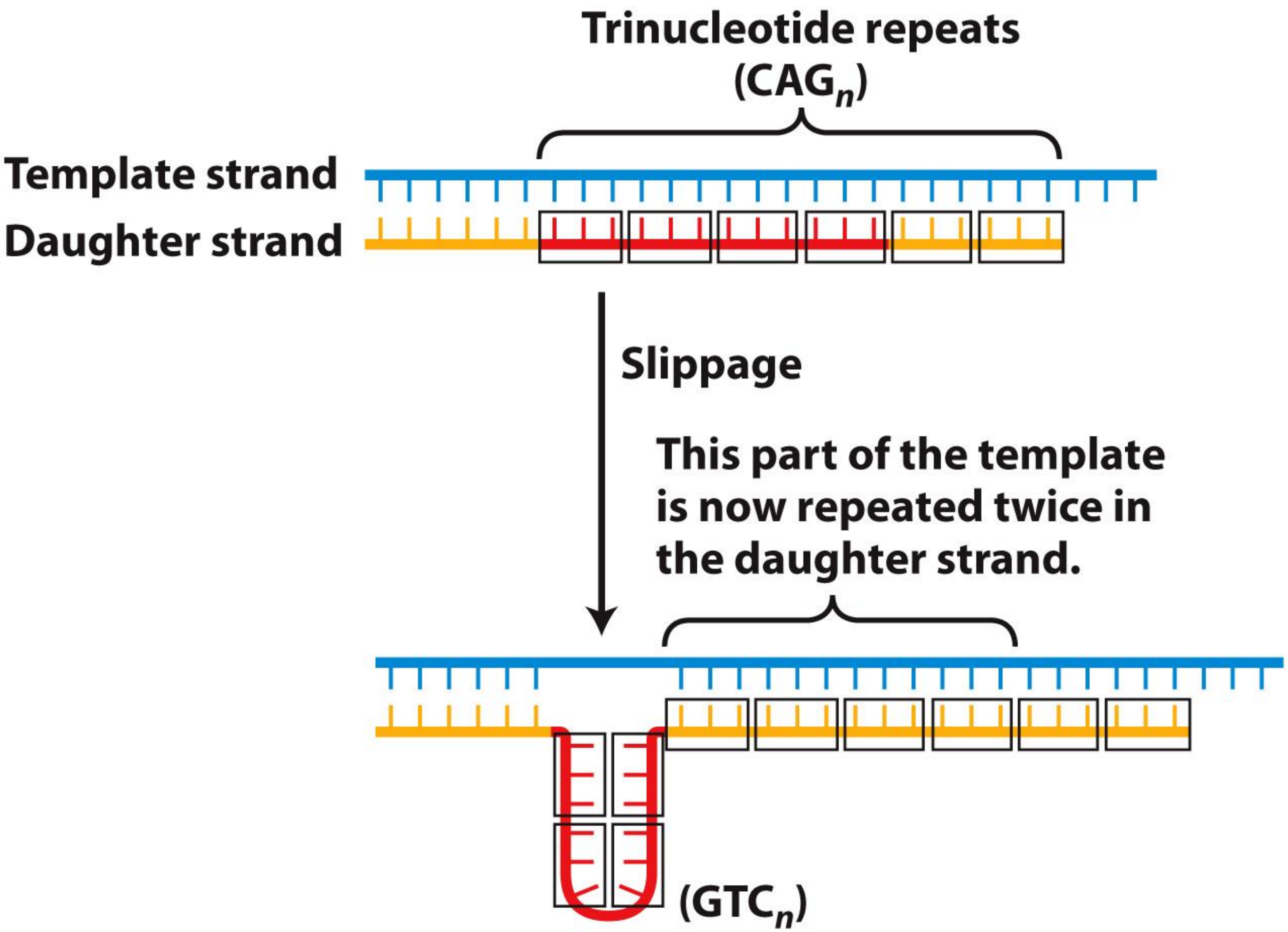


Figure 16-11
Introduction to Genetic Analysis, Eleventh Edition

Strand slippage is the cause of many "trinucleotide repeat diseases"

Here: List of diseases caused by expanded CAG tracts

Disease	Gene	Locus	Protein	CAG repeat size	
				Normal	Disease
Spinobulbar muscular atrophy (Kennedy disease)	*AR*	Xq13–21	Androgen receptor (AR)	9–36	38–62
Huntington's disease	*HD*	4p16.3	Huntingtin	6–35	36–121
Dentatorubral-pallidoluysian atrophy (Haw–River syndrome)	*DRPLA*	12p13.31	Atrophin-1	6–35	49–88
Spinocerebellar ataxia type 1	*SCA1*	6p23	Ataxin-1	6–44[a]	39–82
Spinocerebellar ataxia type 2	*SCA2*	12q24.1	Ataxin-2	15–31	36–63
Spinocerebellar ataxia type 3 (Machado–Joseph disease)	*SCA3* (*MJD1*)	14q32.1	Ataxin-3	12–40	55–84
Spinocerebellar ataxia type 6	*SCA6*	19p13	α_{1A}-voltage-dependent calcium channel subunit	4–18	21–33
Spinocerebellar ataxia type 7	*SCA7*	13p12–13	Ataxin-7	4–35	37–306

Since CAG encodes glutamine (Q), these are known as "polyQ" diseases.

Expanded polyQ tracts lead to abnormal folding, which leads to protein aggregation and neural degeneration.

Hum Mol Genet, Volume 9, Issue 6, 2000, Pages 909–916, https://doi.org/10.1093/hmg/9.6.909

Example: trinucleotide repeat expansion in non-coding region of the FMR1 gene causes Fragile X Syndrome

Methylation of trinucleotide repeats blocks transcription of *FMR1*

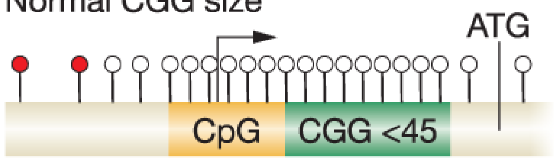
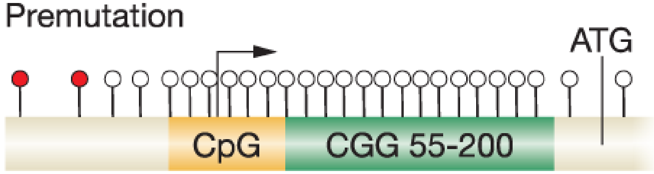
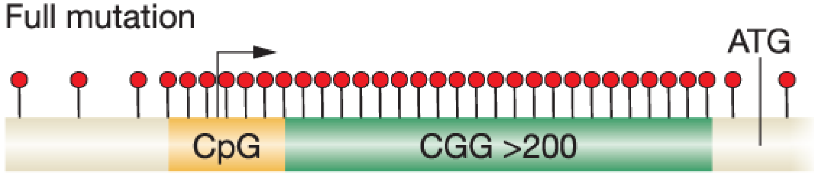
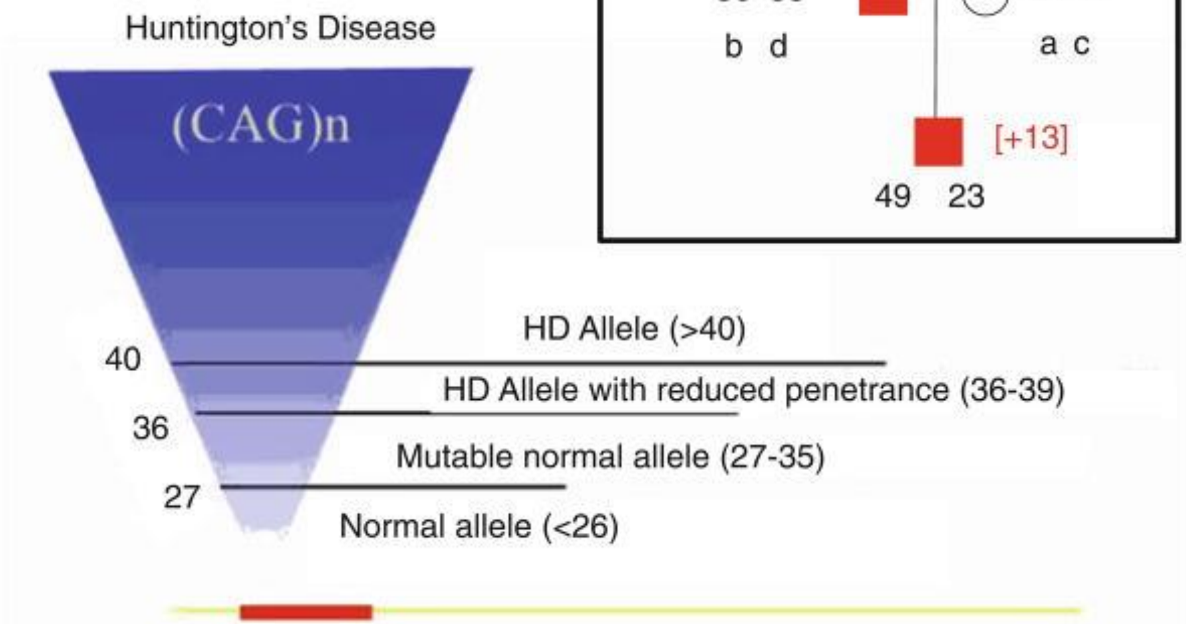
FMR1 gene	Phenotype	Transmission	Methylation	Transcription
Normal CGG size 	Normal	Stable	No	Yes
Premutation 	Largely normal	Unstable, prone to expansion	No	Yes
Full mutation 	Affected	Unstable, prone to expansion	Yes	No

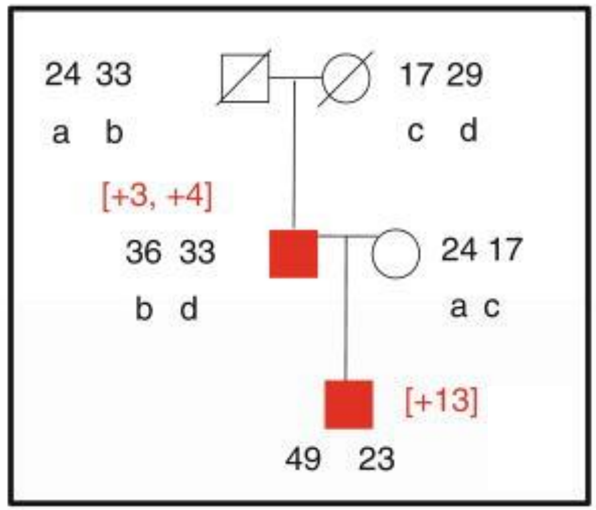
Fig 15.8. The number of trinucleotide CGG repeats in the 5′ UTR of *FMR1* affects phenotype, the probability of repeat expansion, CpG methylation, and transcription. Red and white lollipops indicate methylated and unmethylated cytosines, respectively, CpG indicates a CpG island, the arrow indicates the transcription start site, and ATG indicates the translation start site.

greater n
greater slippage

Example: trinucleotide repeat expansion in coding region of the HTT gene causes Huntington's Disease

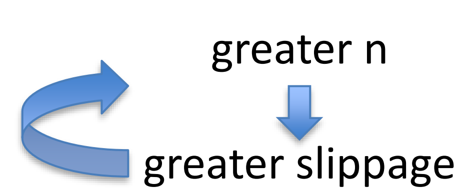
autosomal dominant





Number of repeats in each HTT allele

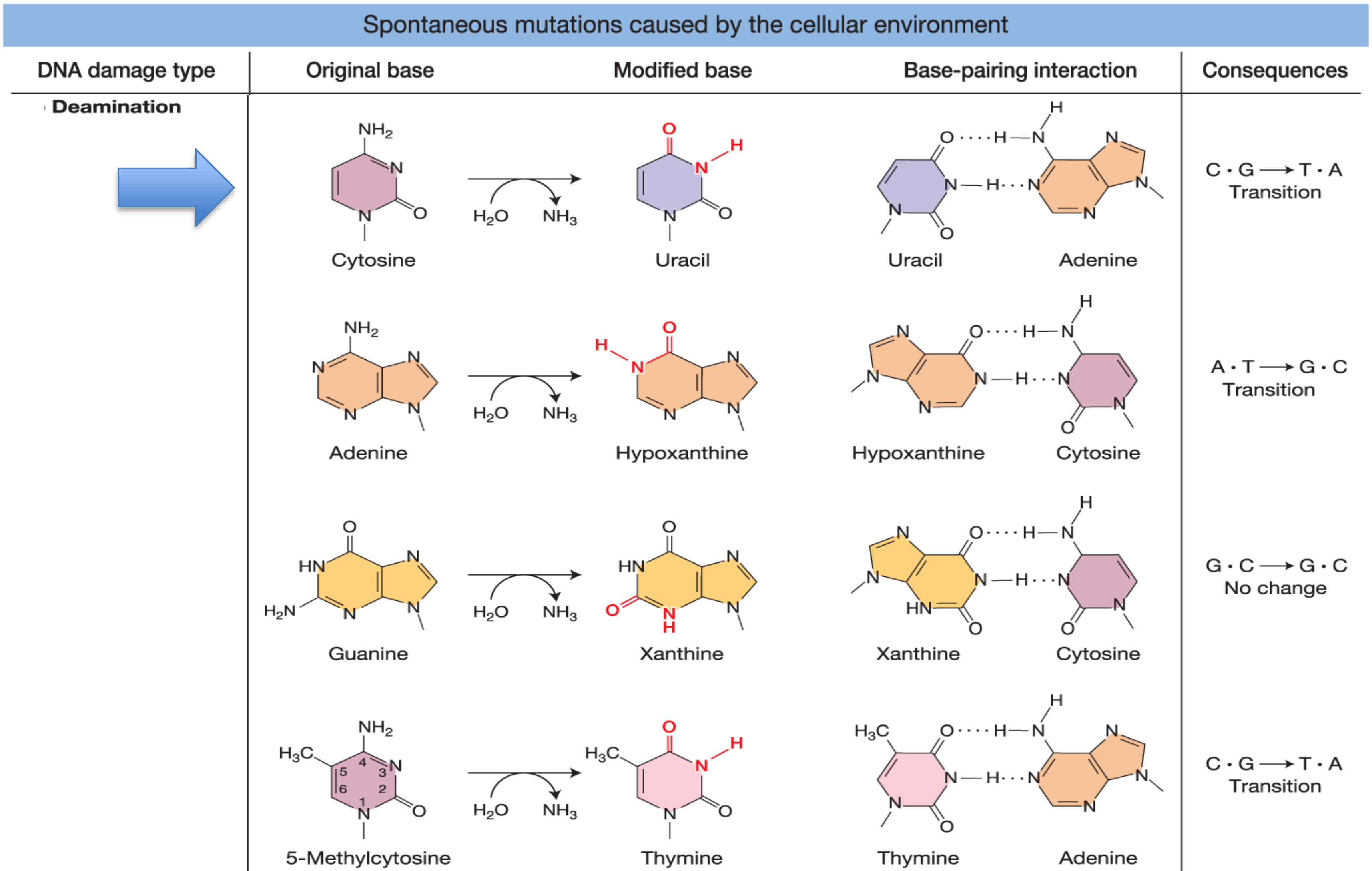
More repeats → earlier age of onset; faster progression of symptoms



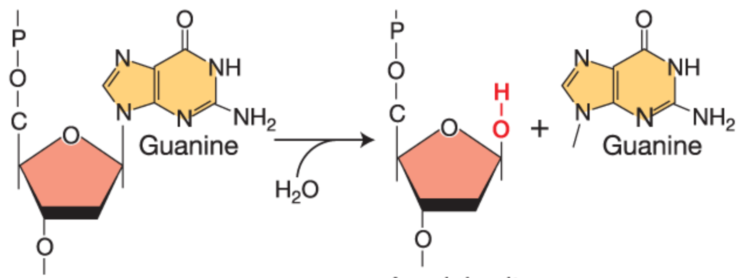
Budworth and McMurray (2013) Methods Mol Biol. 1010: 3-17.

Spontaneous chemical changes to DNA

Spontaneous mutations can be caused by spontaneous chemical changes to DNA



Example of a spontaneous chemical change to DNA: Depurination

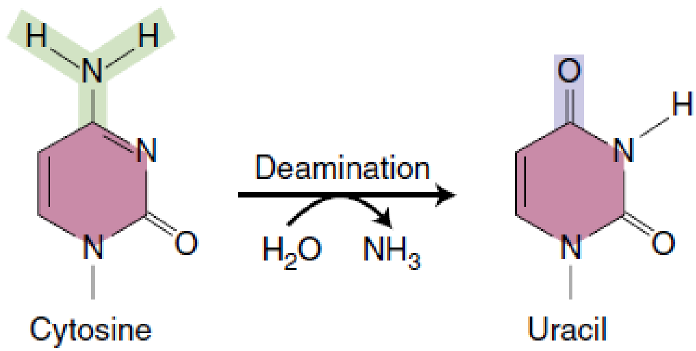
<table>
<tr><th colspan="5">Spontaneous mutations caused by the cellular environment</th></tr>
<tr><th>DNA damage type</th><th>Original base</th><th>Modified base</th><th>Base-pairing interaction</th><th>Consequences</th></tr>
<tr><td>Depurination</td><td colspan="2">
</td><td>None</td><td>Potentially mutagenic

Blocked DNA replication and transcription</td></tr>
</table>

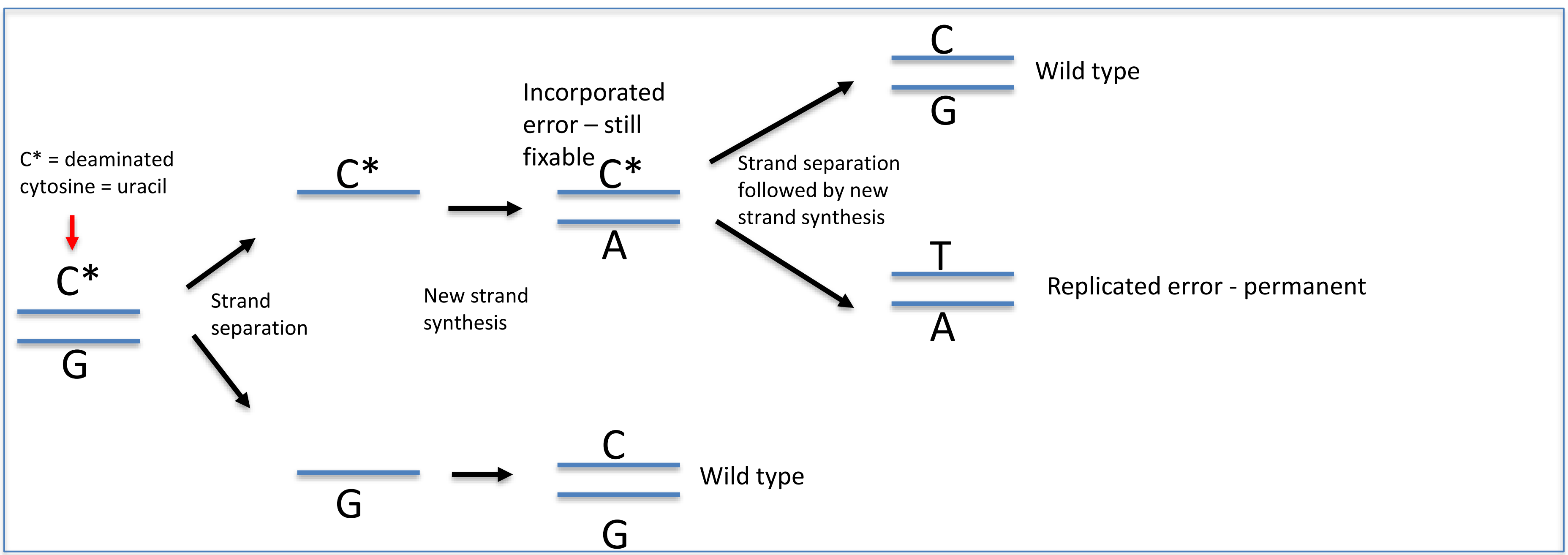
Can lead to sequence errors during replication.
10,000 purines lost per cell in a day! (usually repaired)

Example of a spontaneous chemical change to DNA: Deamination

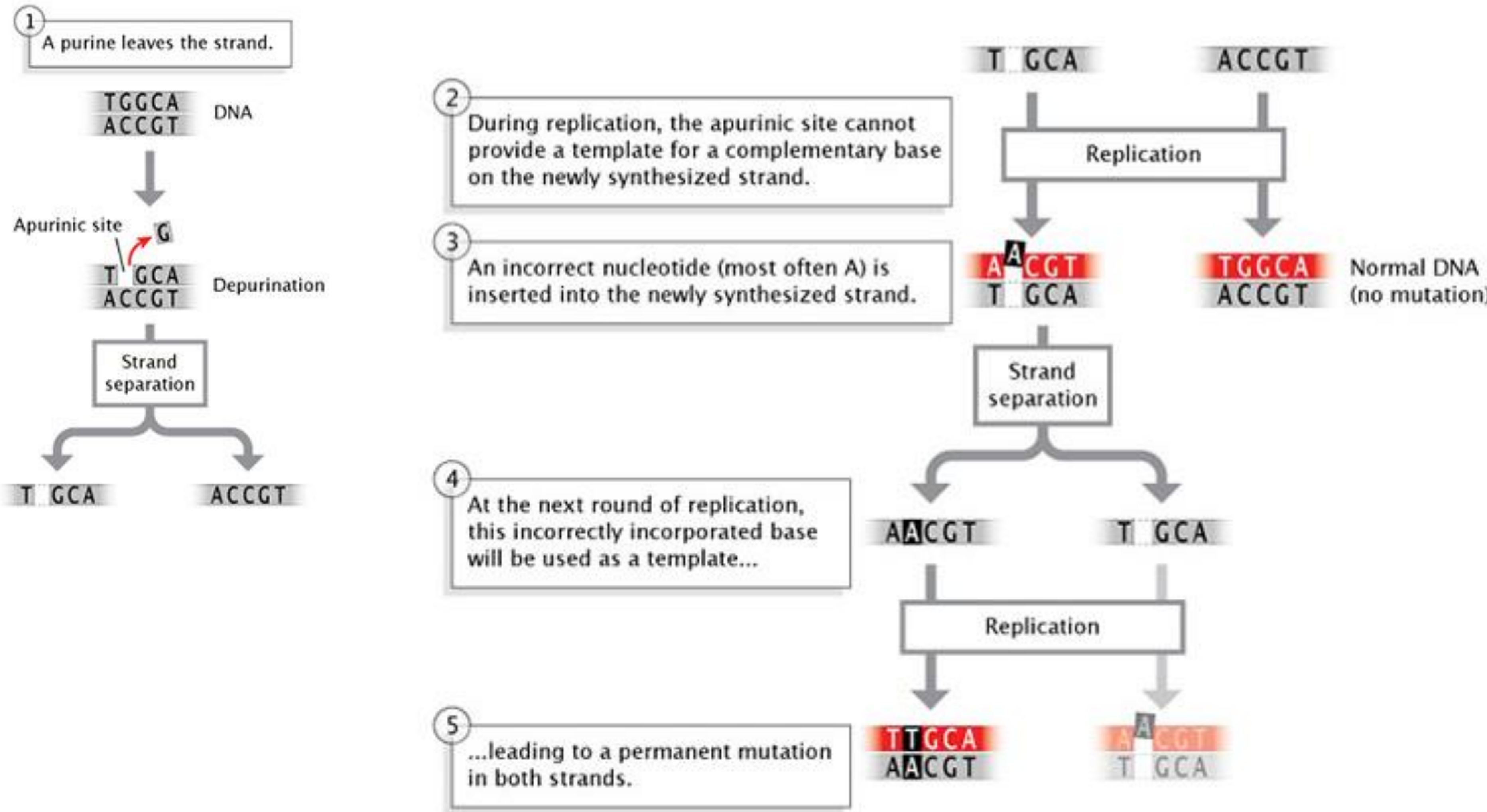
Spontaneous mutations caused by the cellular environment



conversion of a C · G pair into an T · A pair
(C · G → T · A transition)



Mispairing of mutated bases can lead to permanent DNA mutations

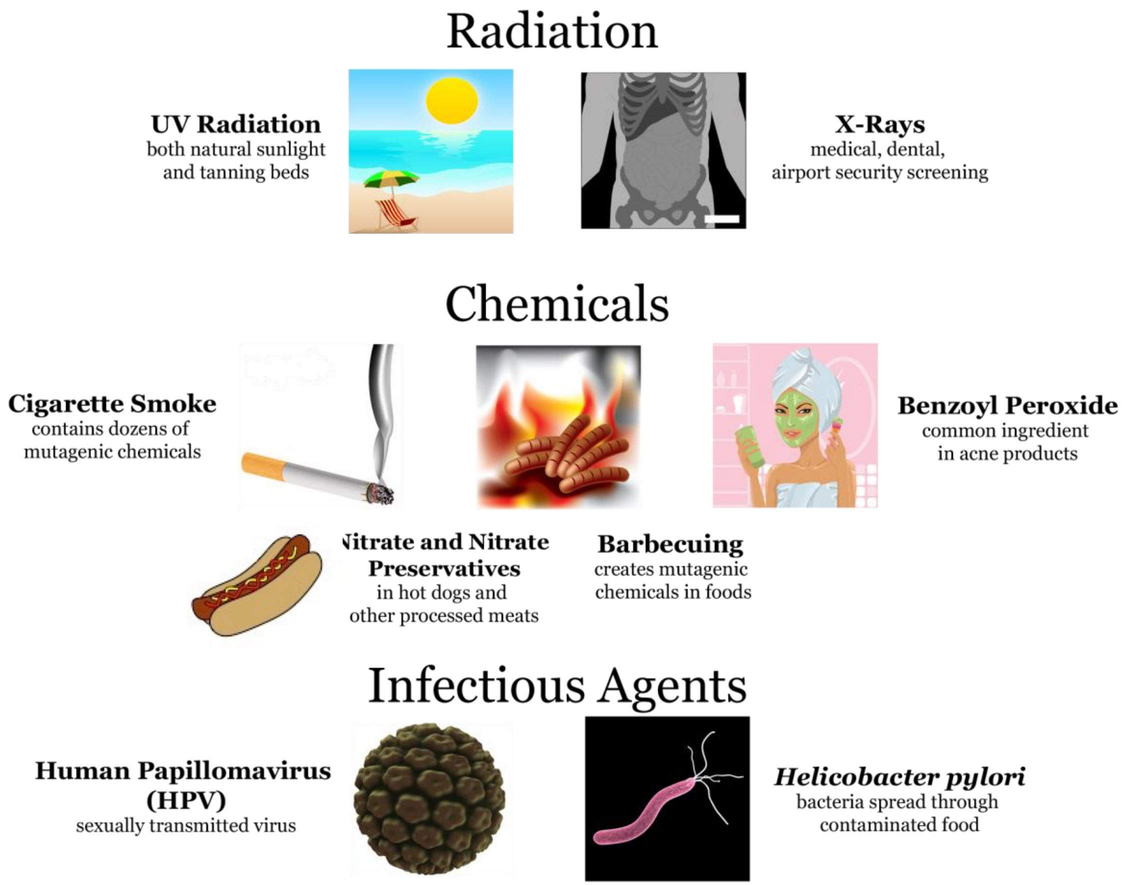


[Genetic Mutation | Learn Science at Scitable (nature.com)]

Molecular basis of **induced** mutations

Molecular basis of **induced** mutations

Mutagens can induce mispairing or damage/non-recognition



Molecular basis of induced mutations

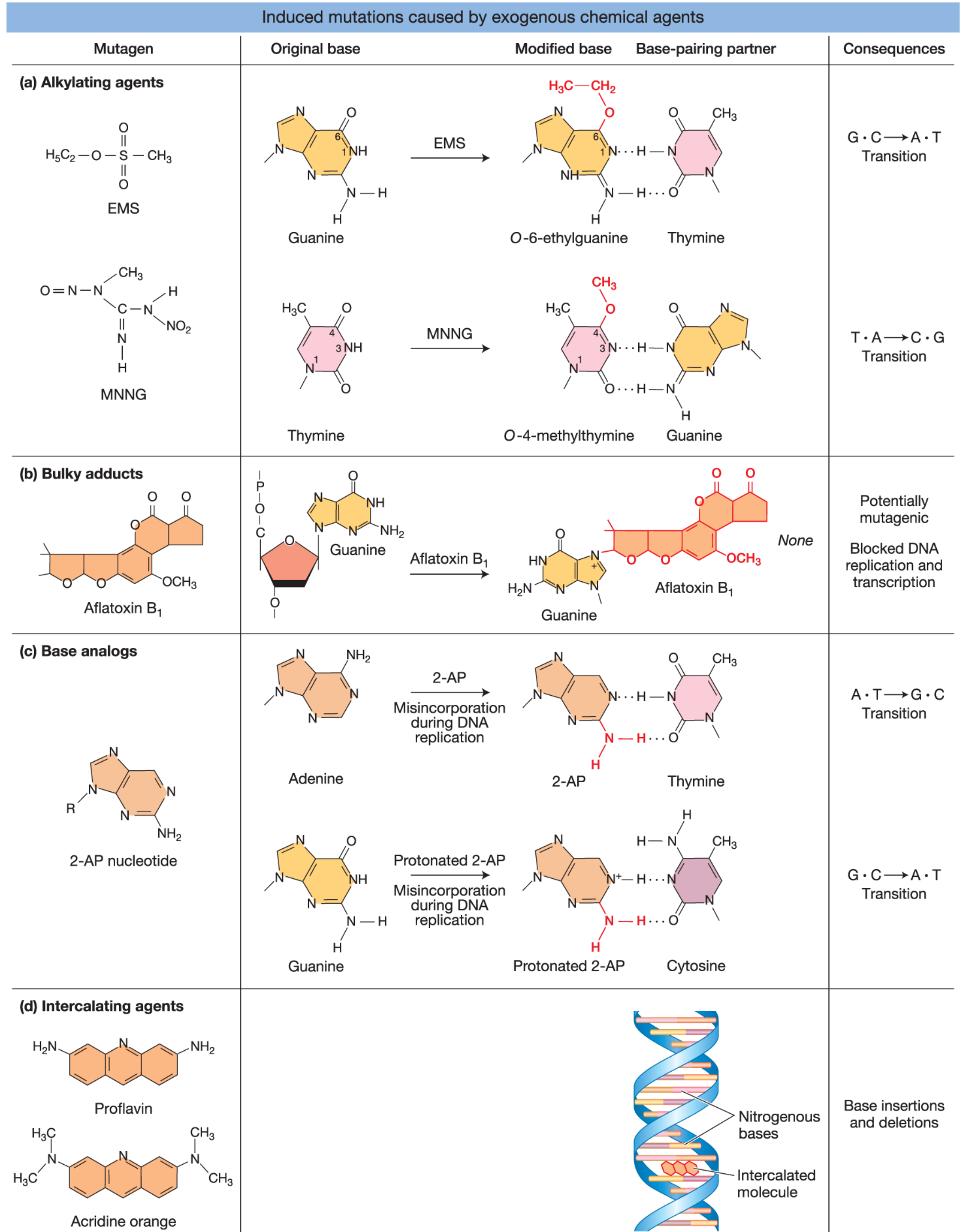
Mutagens can induce mispairing or damage/non-recognition

Mutagens induce mutations by at least three different mechanisms.

- *replace* a base in DNA, i.e. get incorporated instead of an actual base (base analogs)
- *alter* a base so that it mispairs with another base (e.g. alkylating agents)
- *damage* a base so that it can no longer base pair with any base (e.g. UV light, ionizing radiation)

Molecular basis of induced mutations

Some examples of mutagens and their effects



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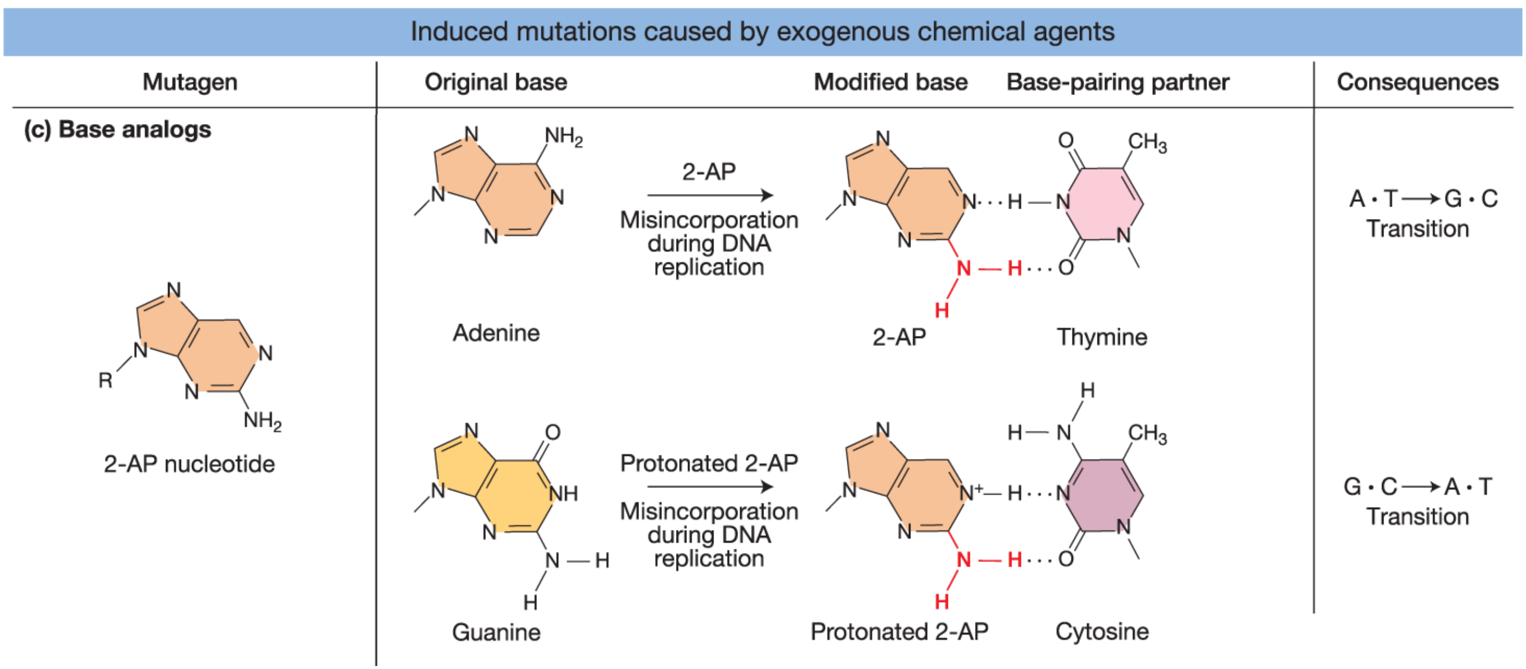
Molecular basis of induced mutations

Base replacement – base analogs

- Can be incorporated into DNA in place of normal bases
- Can be less stable, shifting their base-pairing affinities and leading to nucleotide changes.
- Example: 2-aminopurine is an analog of A, but when protonated can mispair with C, leading to transitions



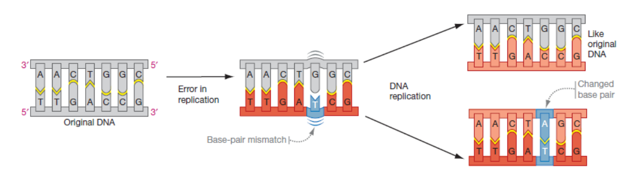
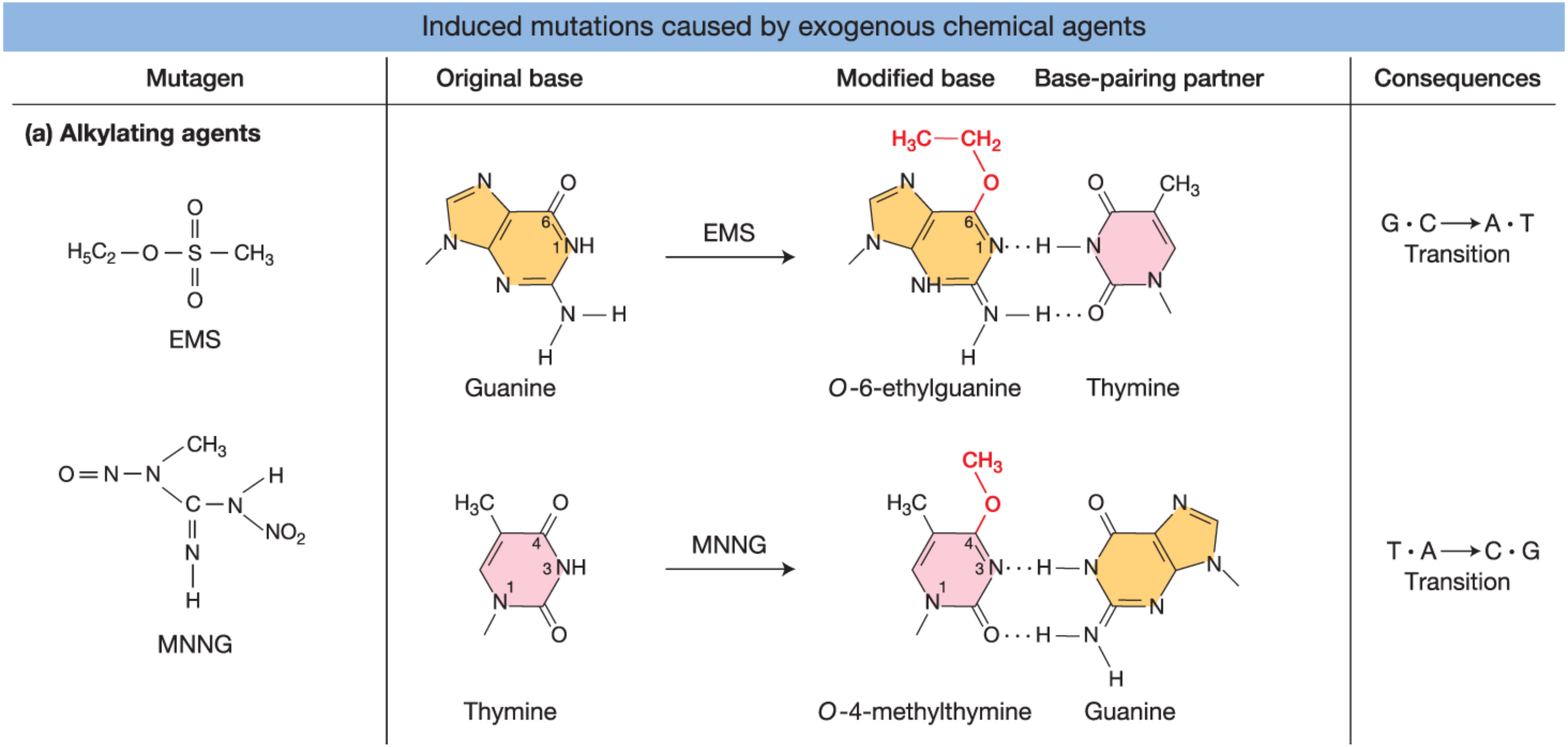


FIGURE 16.8 Unrepaired Mistakes in DNA Synthesis Lead to Point Mutations.

Molecular basis of induced mutations

Base alteration – alkylating agents



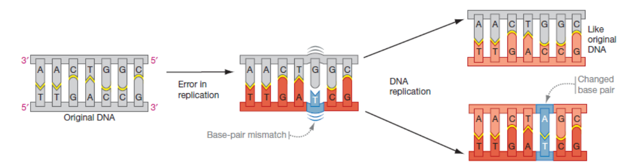
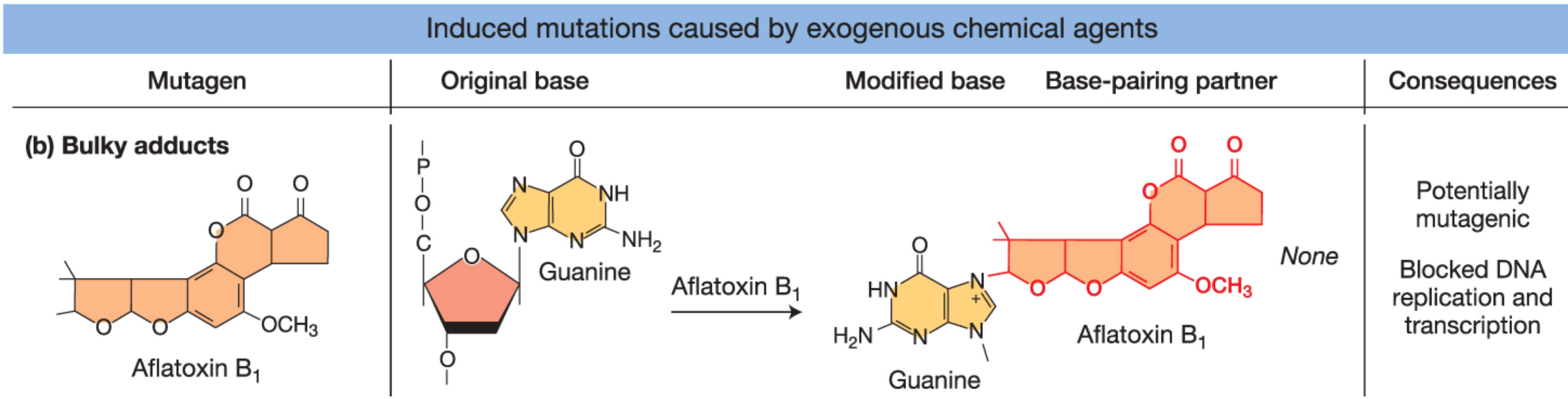


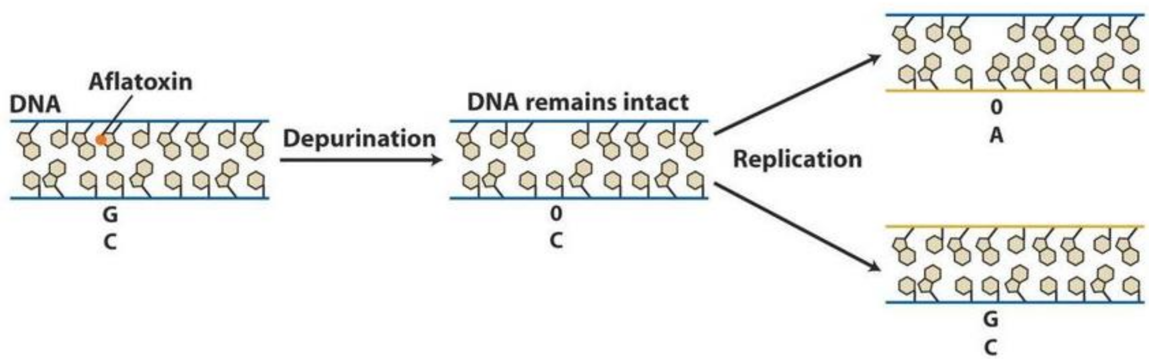
FIGURE 16.8 Unrepaired Mistakes in DNA Synthesis Lead to Point Mutations.

Molecular basis of induced mutations

Base damage – bulky adducts

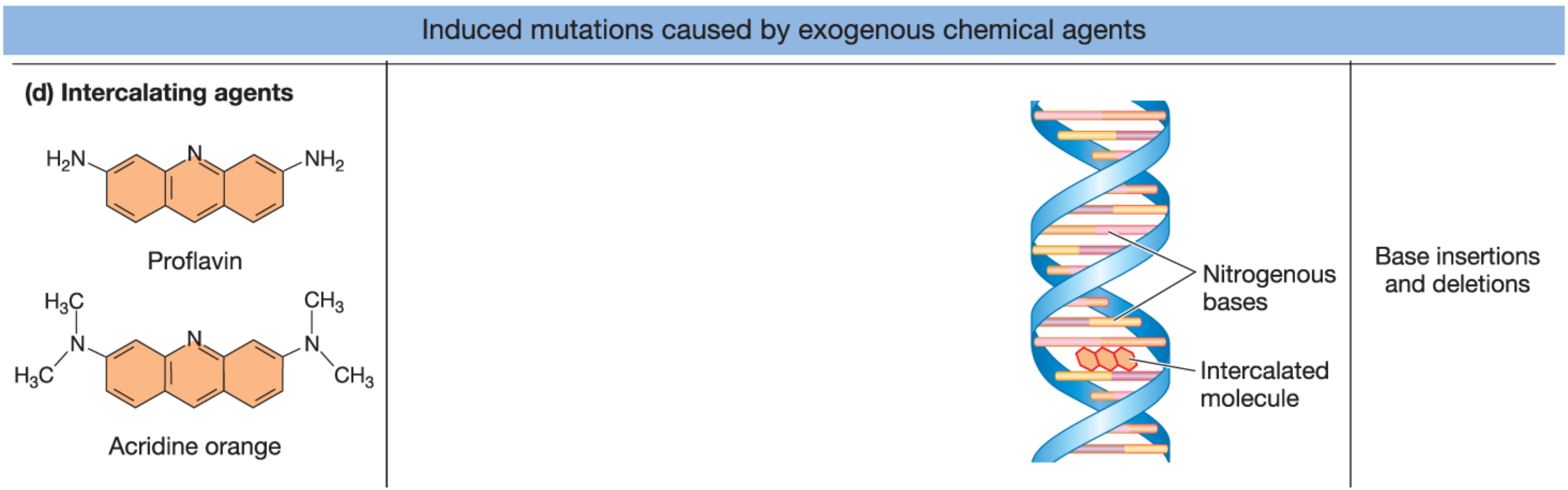


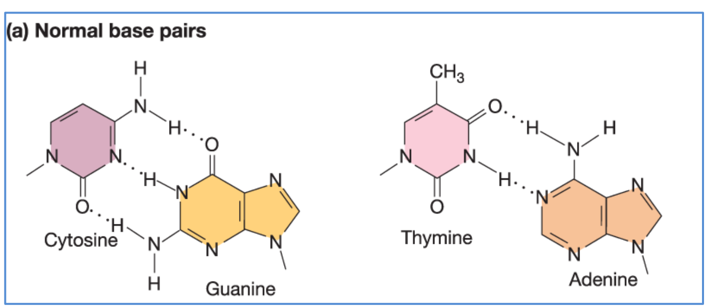
- Example: Aflatoxin B1 (from Aspergillus (fungus)-infected peanuts)
- Attaches to N-7 of guanine, disrupts sugar-base bond to generate an apurinic site
- In this example, this damage results in a GC→TA transversion
- All compounds of this class induce mutations, although the mechanisms are not always clear.



Molecular basis of induced mutations

Intercalating agents

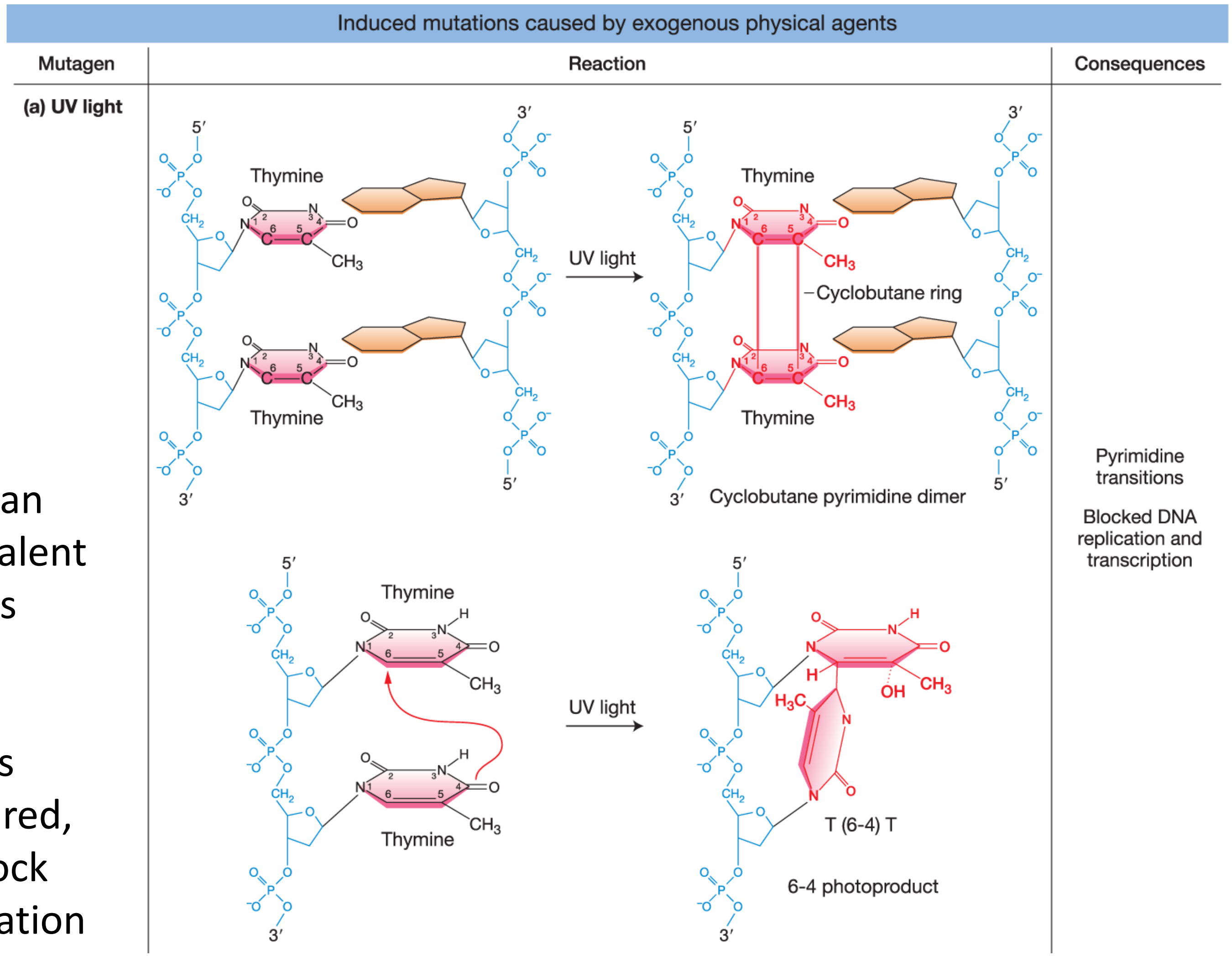




- Mimic base pairs and insert into DNA double helix
- Disrupt spacing and <u>can cause insertions or deletions of single nucleotide pairs</u>

Molecular basis of induced mutations

Base damage – UV light



- UVB light can induce covalent interactions between adjacent pyrimidines
- If not repaired, this can block DNA replication

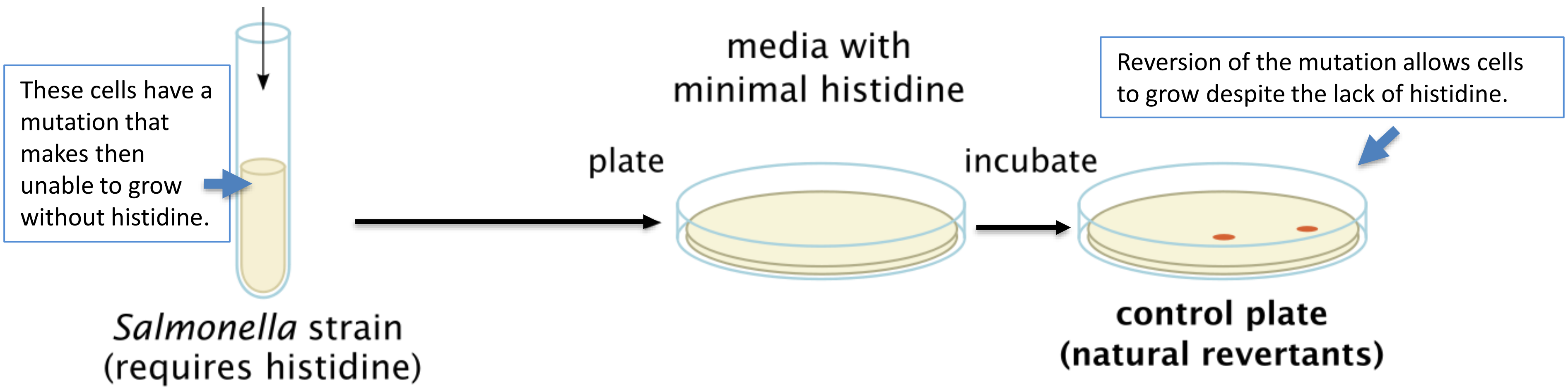
How do we know if something is a mutagen?

How do we know if something is a mutagen?

Ask if it can cause mutations! But how?

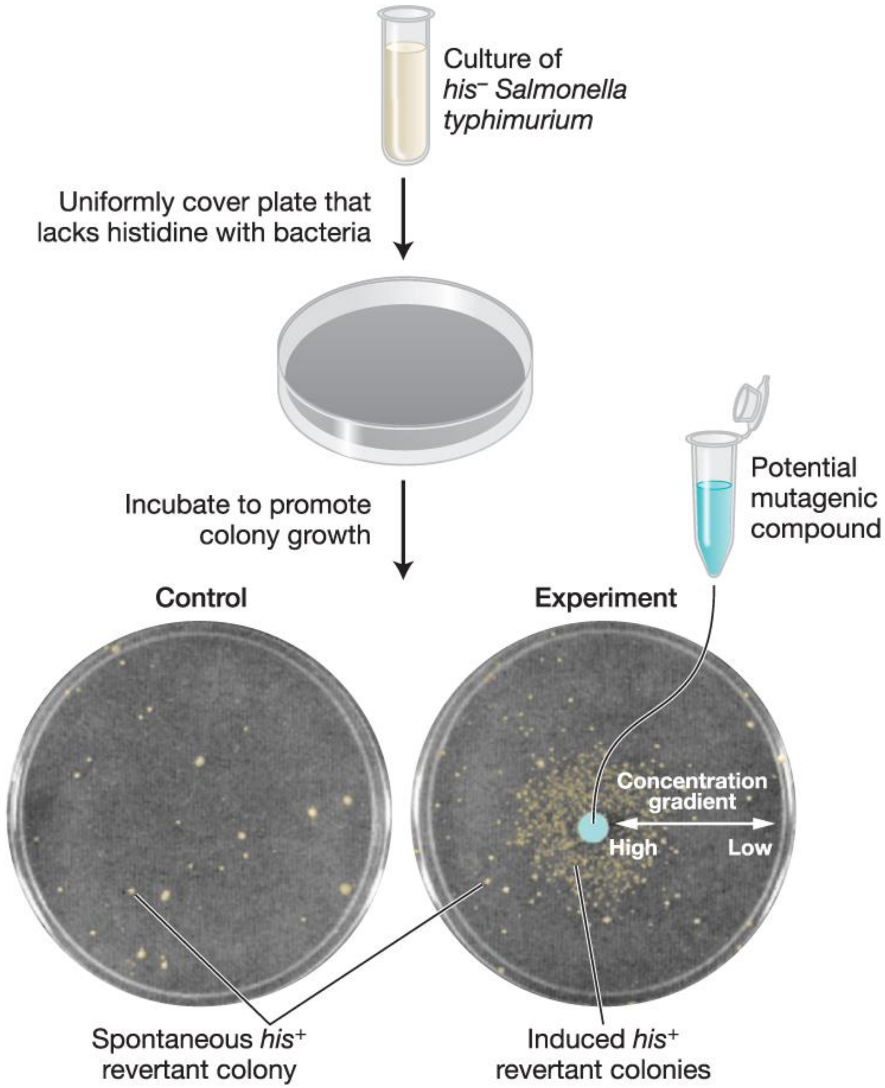
The "Ames test"

Starting point: a *his*-mutant strain of Salmonella bacteria, which cannot grow in the absence of histidine. Growth in the presence of only low levels of histidine means that the bacterium has undergone a spontaneous mutation that has allowed it to survive. These rare surviving colonies are called "revertants."



Rationale behind the Ames test: if treatment of these *his*-mutant Salmonella cells with a particular compound increases the number of revertants, then that compound is a mutagen.

The Ames test for mutagenic compounds



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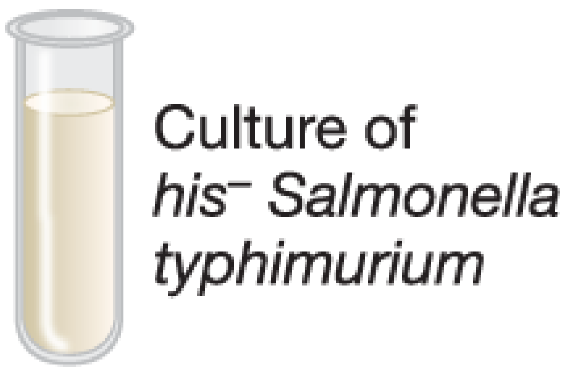
Republished with permission of Elsevier, Ames BN, McCann J, Yamasaki E., "Methods for detecting carcinogens and mutagens with the Salmonella*/mammalian-microsome mutagenicity test,"* Mutation Research*, 1975, December; 31 (6):347–64, Figure 2. Permission conveyed through Copyright Clearance Center, Inc.*

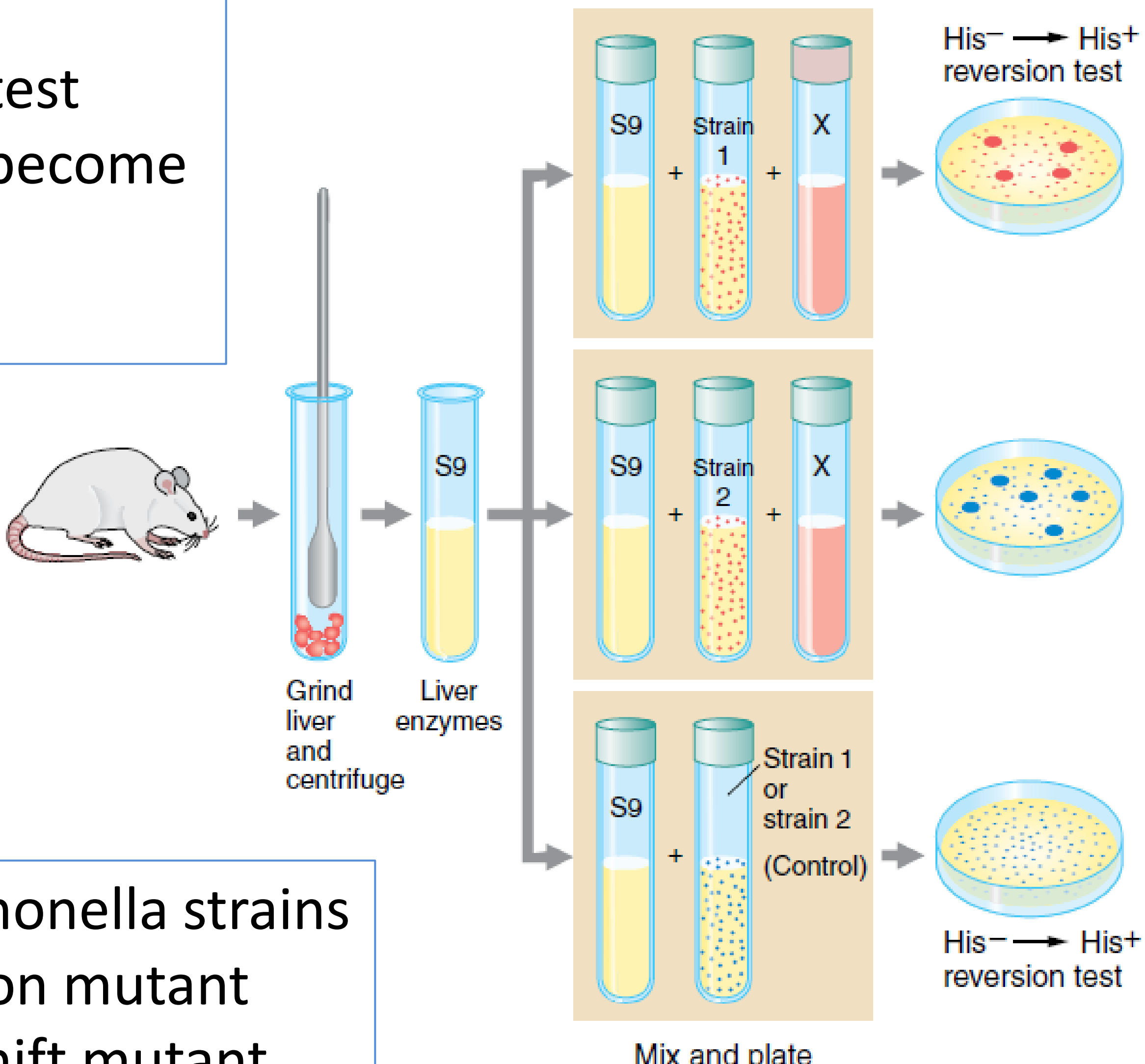
A problem with the Ames test



≠

The Ames test for mutagenic compounds

Revised Ames test:
Add liver enzymes to test chemicals that might become mutagenic only when metabolized



Why?

use two different Salmonella strains
Strain 1 = His- transition mutant
Strain 2 = His- frameshift mutant

Think Break

Reflection

Why did the revised Ames test include these two different Salmonella strains?

		Single base substitution			
	No mutation	Silent mutation (synonymous)	Missense mutation (nonsynonymous)		Nonsense mutation
			Conservative	Nonconservative	
DNA	CAT AAG CAG AGT ACT GTA TTC GTC TCA TGA	CAT AAA CAG AGT ACT GTA TTT GTC TCA TGA	CAT AGG CAG AGT ACT GTA TCC GTC TCA TGA	CAT ACG CAG AGT ACT GTA TGC GTC TCA TGA	CAT TAG CAG AGT ACT GTA ATC GTC TCA TGA
mRNA	CAU AAG CAG AGU ACU	CAU AAA CAG AGU ACU	CAU AGG CAG AGU ACU	CAU ACG CAG AGU ACU	CAU UAG CAG AGU ACU
Protein	His Lys Gln Ser Thr	His Lys Gln Ser Thr	His Arg Gln Ser Thr	His Thr Gln Ser Thr	His Stop

		Single base insertion or base deletion (indel)	
	No mutation	Insertion (frameshift)	Deletion (frameshift)
DNA	CAT TGC GAC AAG GAT AGT ACT CCT GTA ACG CTG TTC CTA TCA TGA GGA	CAT GTG CGA CAA GGA TAG TAC TCC T GTA CAC GCT GTT CCT ATC ATG AGG A	T CAT GCG ACA AGG ATA GTA CTC CT GTA CGC TGT TCC TAT CAT GAG GA A
mRNA	CAU UGC GAC AAG GAU AGU ACU CCU	CAU GUG CGA CAA GGA UAG UAC UCC U	CAU GCG ACA AGG AUA GUA CUC CU
Protein	His Cys Asp Lys Val Ser Thr Pro	His Val Arg Gln Gly Stop	His Ala Thr Arg Ile Val Leu

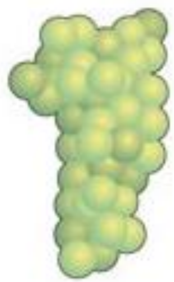
Bioinformatics and Genomics

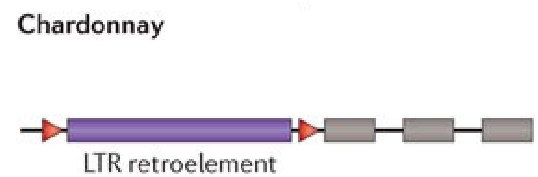
Bioinformatics and genome annotation

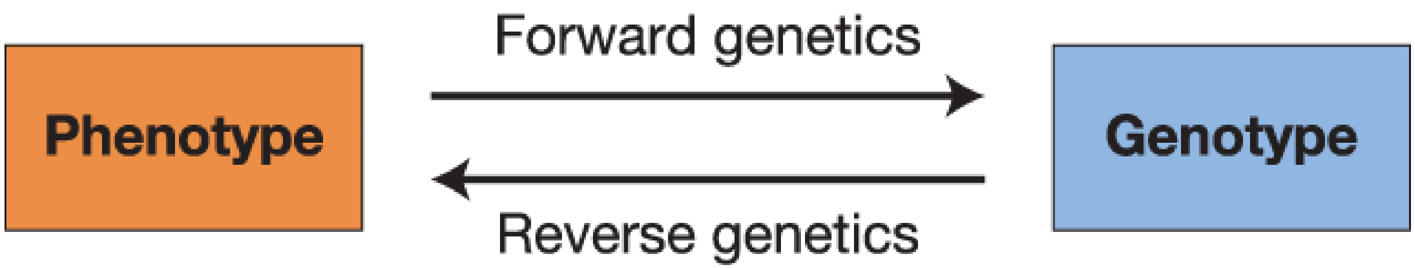
Comparative genomics

Functional genomics and reverse genetics

How can we read and understand a genome?





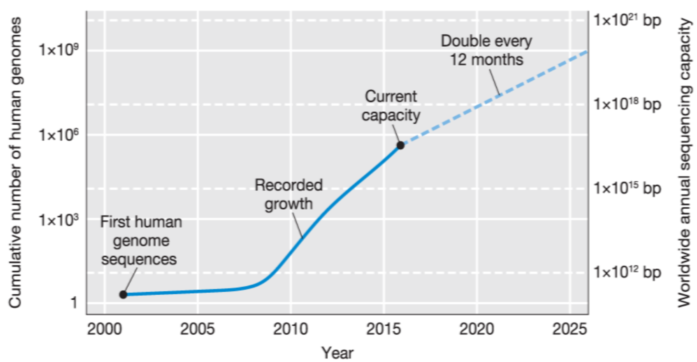


Reading the Genome

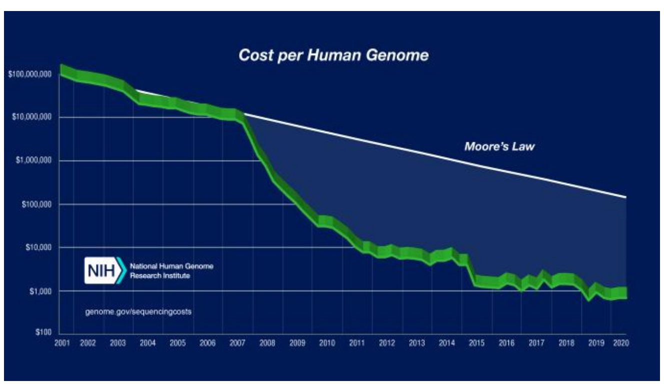
…begins with sequencing

Improvements in sequencing technology and costs have led to the generation of a lot of genome sequence

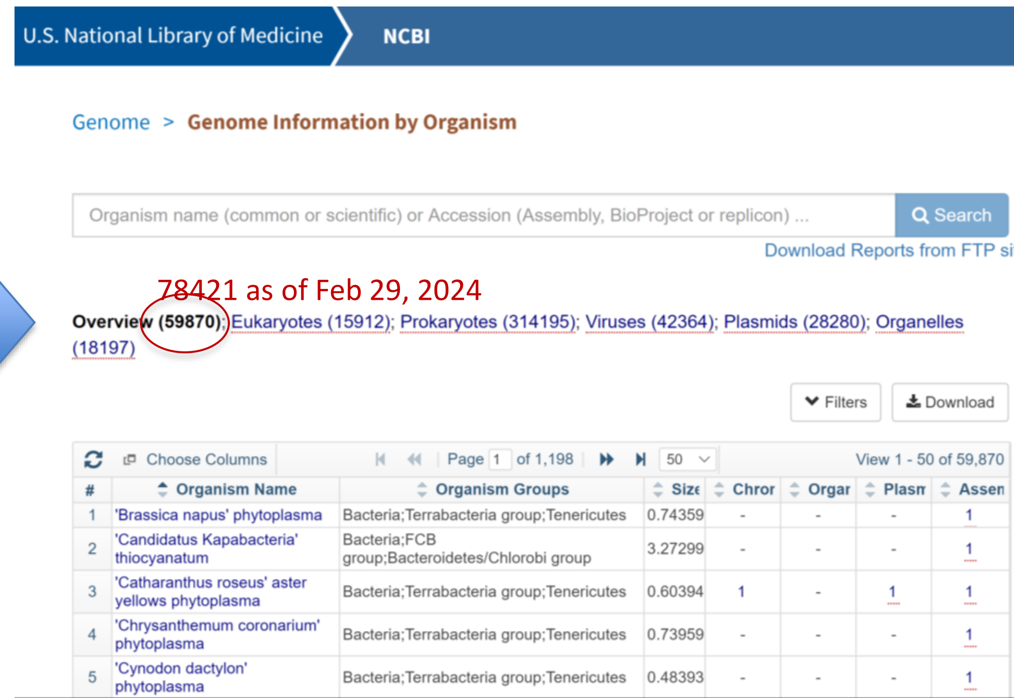
Growth of DNA sequencing capacity



Decrease in DNA sequencing costs



A lot of genomes have been sequenced


U.S. National Library of Medicine
NCBI

Genome > Genome Information by Organism

Organism name (common or scientific) or Accession (Assembly, BioProject or replicon) ... Search

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Overview (59870); Eukaryotes (15912); Prokaryotes (314195); Viruses (42364); Plasmids (28280); Organelles (18197)

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#	Organism Name	Organism Groups	Size	Chror	Orgar	Plasm	Assen
1	'Brassica napus' phytoplasma	Bacteria;Terrabacteria group;Tenericutes	0.74359	-	-	-	1
2	'Candidatus Kapabacteria' thiocyanatum	Bacteria;FCB group;Bacteroidetes/Chlorobi group	3.27299	-	-	-	1
3	'Catharanthus roseus' aster yellows phytoplasma	Bacteria;Terrabacteria group;Tenericutes	0.60394	1	-	1	1
4	'Chrysanthemum coronarium' phytoplasma	Bacteria;Terrabacteria group;Tenericutes	0.73959	-	-	-	1
5	'Cynodon dactylon' phytoplasma	Bacteria;Terrabacteria group;Tenericutes	0.48393	-	-	-	1

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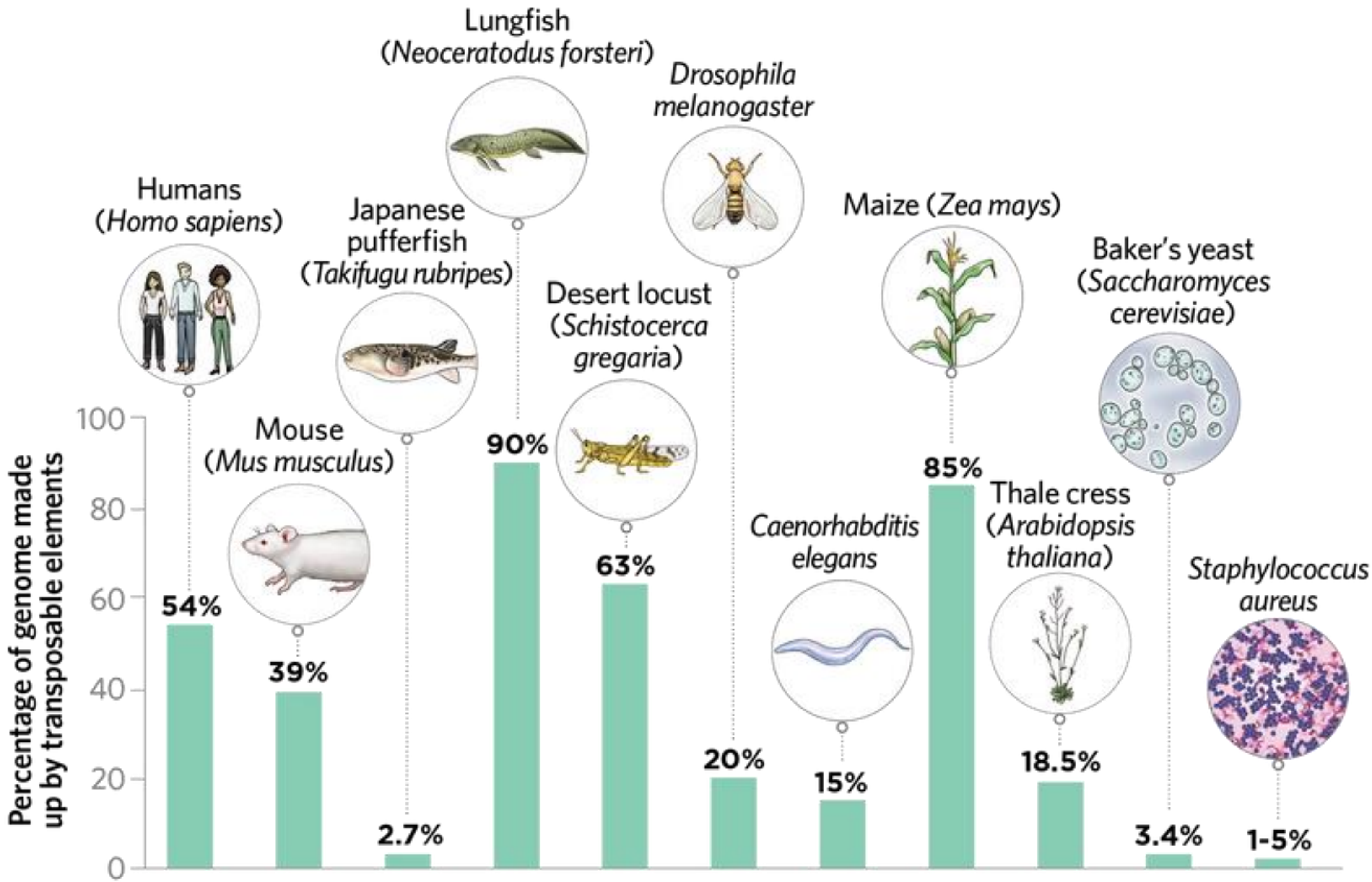
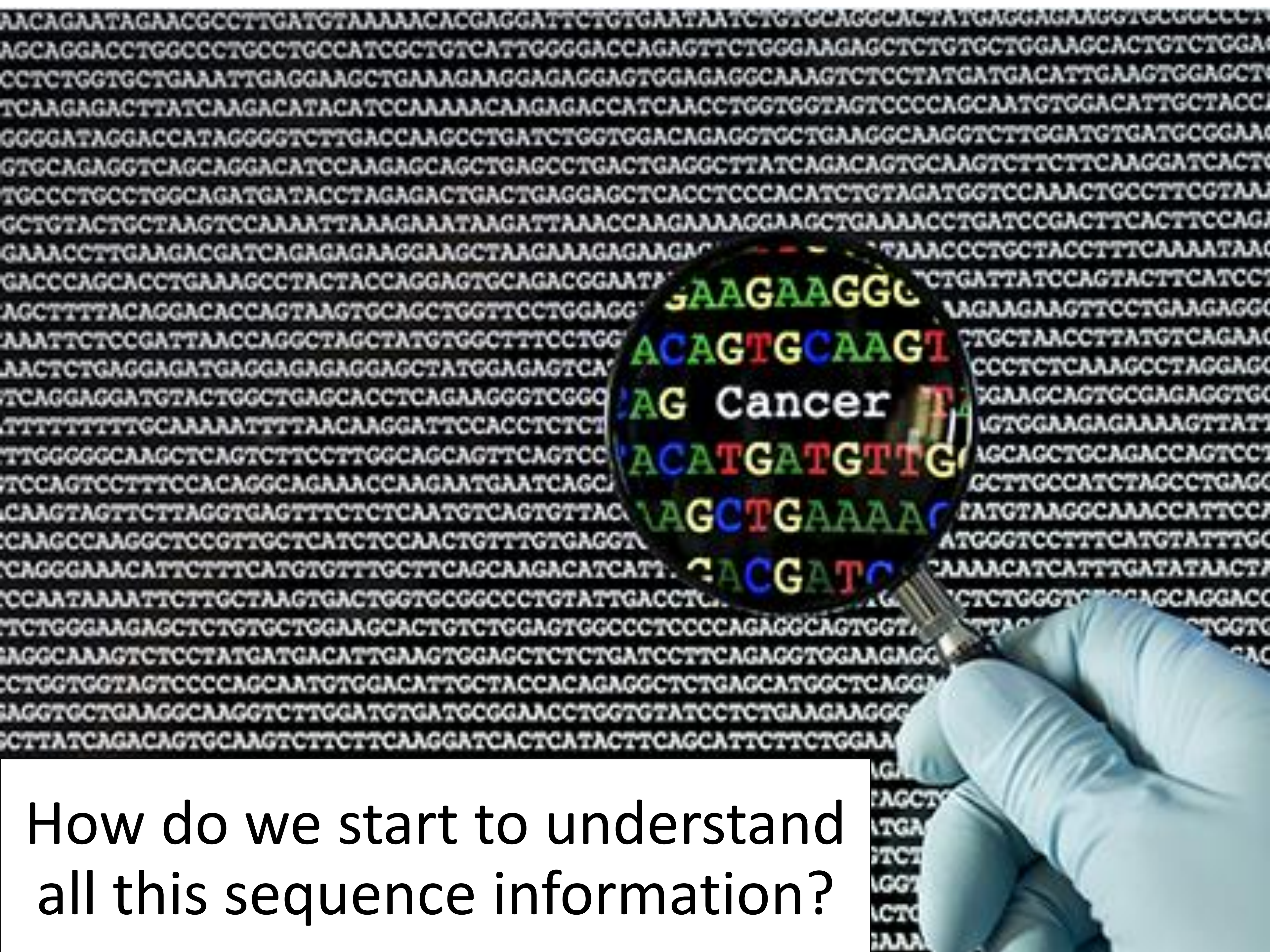
Improvements in sequencing repetitive DNA reveal TE abundance



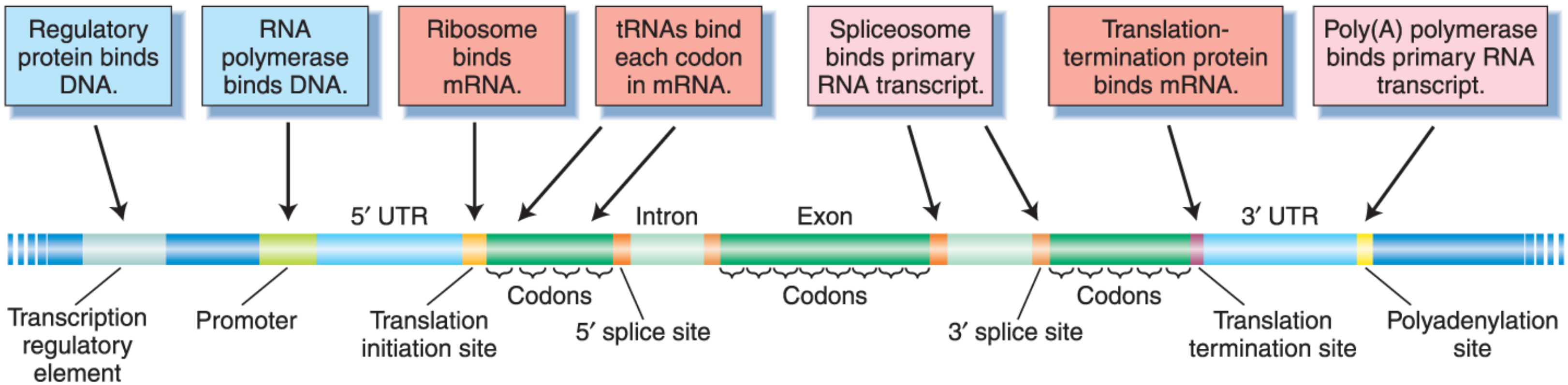
[Adapting with a Little Help from Jumping Genes | TS Digest | The Scientist (the-scientist.com)]



How do we start to understand all this sequence information?

Start with what we know best

Look for sequences associated with protein-encoding genes



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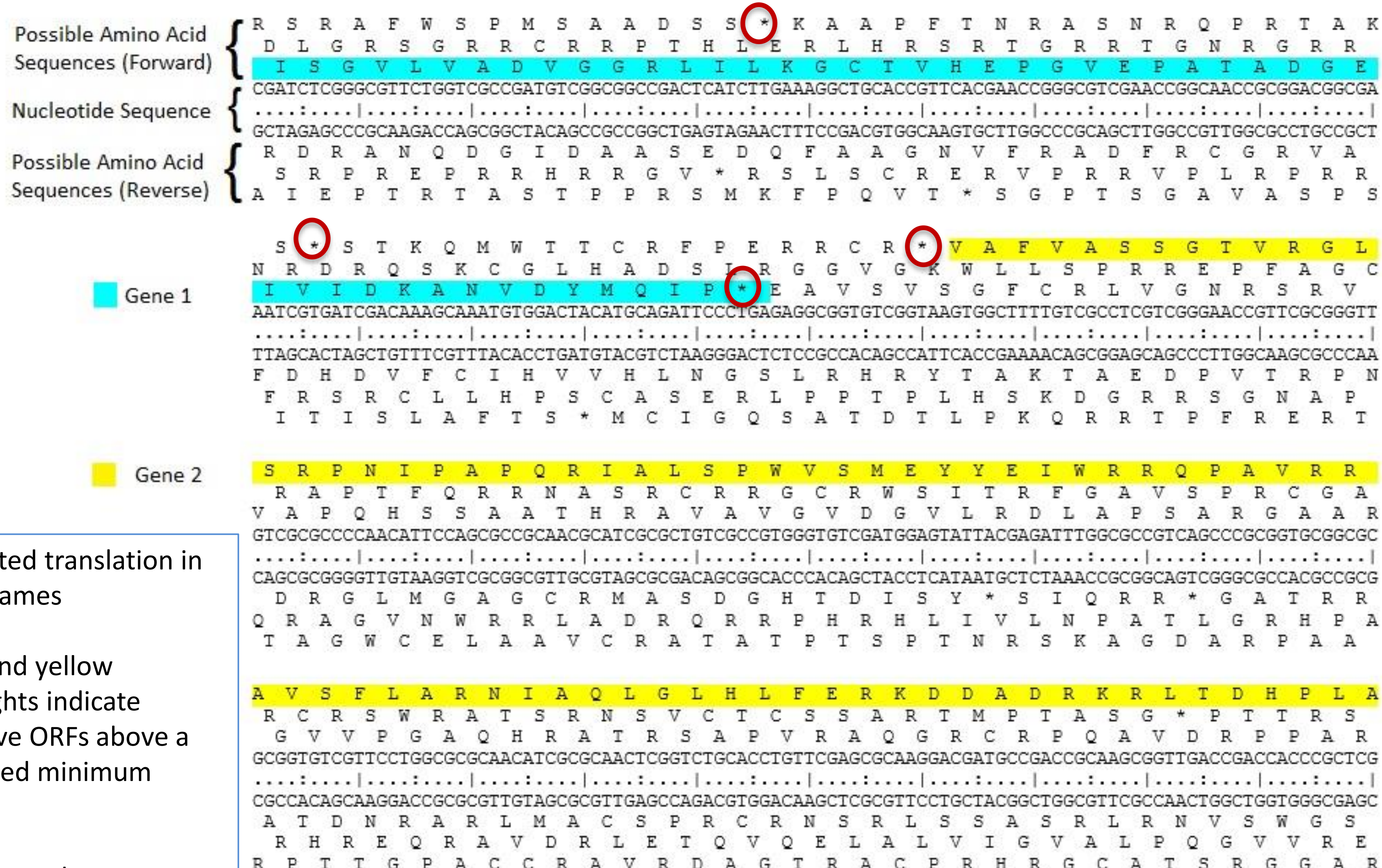
First step: look for open-reading frames ("ORFs,"i.e. stretches of codons that aren't stop codons) as evidence of possible exons/genes.

Look for open reading frames (ORFs) as evidence of possible exons



Predicted translation in all 6 frames

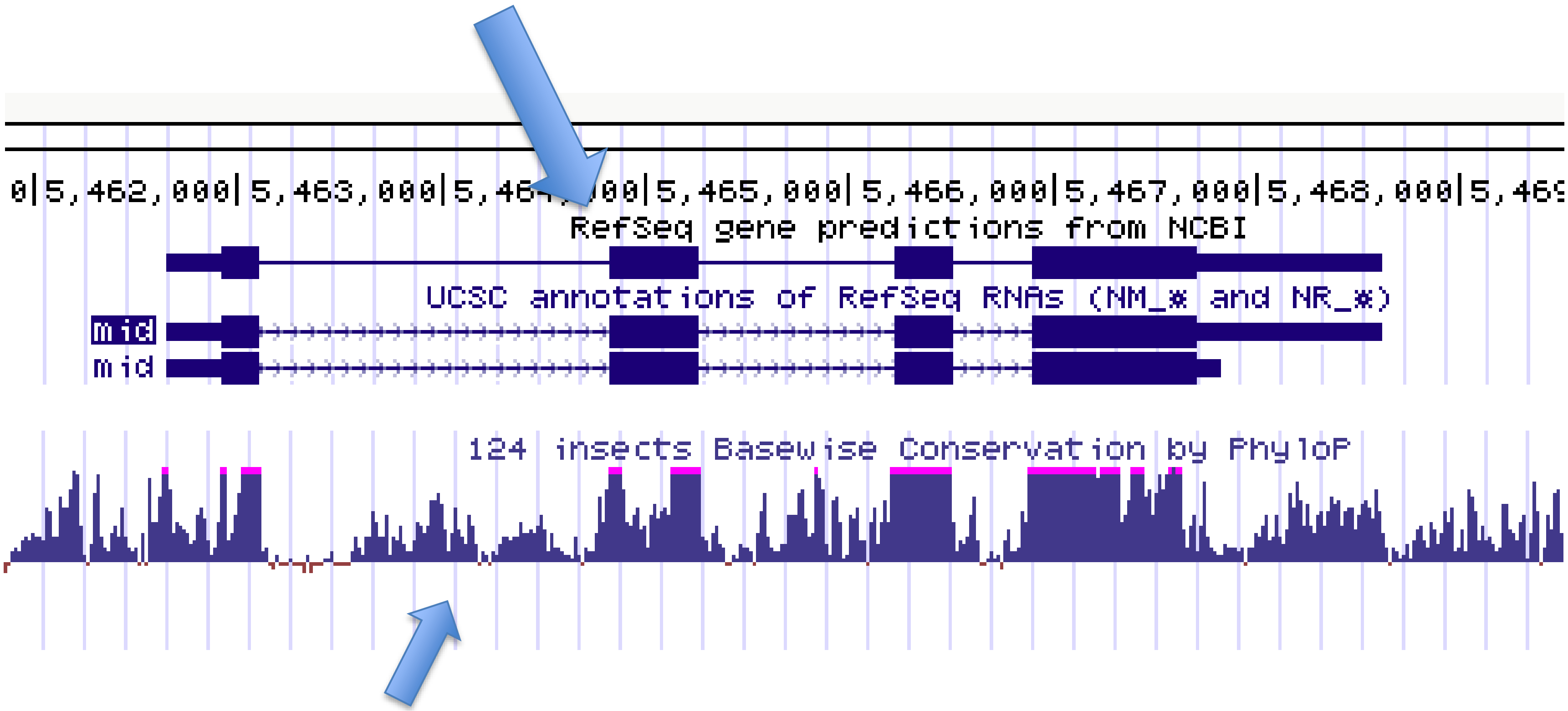
Blue and yellow highlights indicate putative ORFs above a specified minimum length

* = stop codon

[Open reading frame - Wikipedia]

Check for conservation of a predicted ORF across species as evidence that it is part of a real gene.

predicted open reading frames



sequence conservation across 100+ insect species

peak conservation aligns with ORFs → suggests that they are actual exons of a real gene

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Check for "codon bias" as evidence that an ORF is part of a real gene.

Genetic code: same for all organisms

Amino Acids			Codons					
Alanine	Ala	A	GCA	GCC	GCG	GCU		
Cysteine	Cys	C	UGC	UGU				
Aspartic acid	Asp	D	GAC	GAU	GAC	GAU		
Glutamic acid	Glu	E	GAA	GAG				
Phenylalanine	Phe	F	UUC	UUU				
Clycine	Gly	G	GGA	GCG	GGG	GGU		
Histidine	His	H	CAC	CAU				
Isoleucine	Ile	I	AUA	AUC	AUU			
Lysine	Lys	K	AAA	AAG				
Leucine	Leu	L	UUA	UUG	CUA	CUC	CUG	GUU
Methionine	Met	M	AUG					
Asparagine	Asn	N	AAC	AAU				
Proline	Pro	P	CCA	CCC	CCG	CCU		
Glutamine	Gln	Q	CAA	CAG				
Arginine	Arg	R	AGA	AGG	CGA	CGC	CGG	CGU
Serine	Ser	S	AGC	AGU	UCA	UCC	UCG	UCU
Threonine	Thr	T	ACA	ACC	ACG	ACU		
Valine	Val	V	GUA	GUC	GUG	GUU		
Tryptophan	Trp	W	UGG					
Tyrosine	Tyr	Y	UAC	UAU				

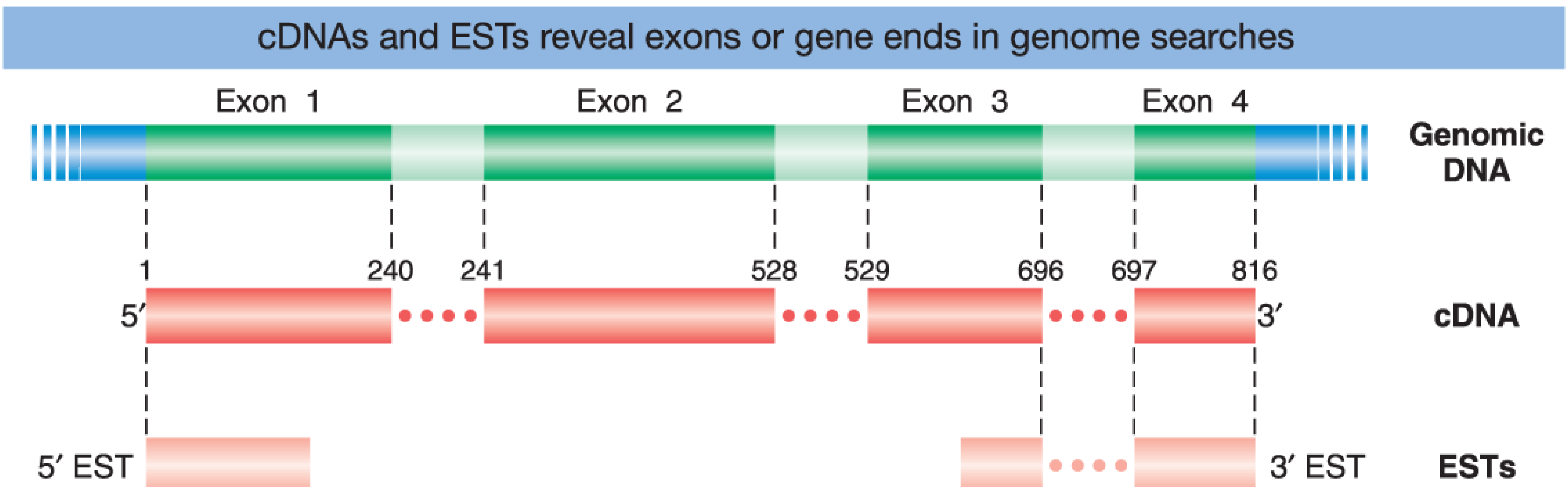
Codon usage: can differ across species

Codon	Human	Drosophila	E. coli
Arginine:			
AGA	22 %	10%	1%
AGG	23 %	6%	1%
CGA	10 %	8%	4 %
CGC	22 %	49%	39 %
CGG	14 %	9%	4 %
CGU	9 %	18%	49%
Total number of arginine codons	2403	506	149
Total number of genes	195	46	149

Most amino acids can be coded for by more than one codon, but different species have different frequencies of synonymous codon usage. This codon preference reflects relative tRNA abundance and can be a signature of an ORF that's part of a real gene.

Look for cDNAs that map to the putative gene

Identification of a cDNA that maps to a predicted ORF provides the best evidence that the ORF is part of a real gene.



An EST ("expressed sequence tag") is a cDNA that has not been completely sequenced.

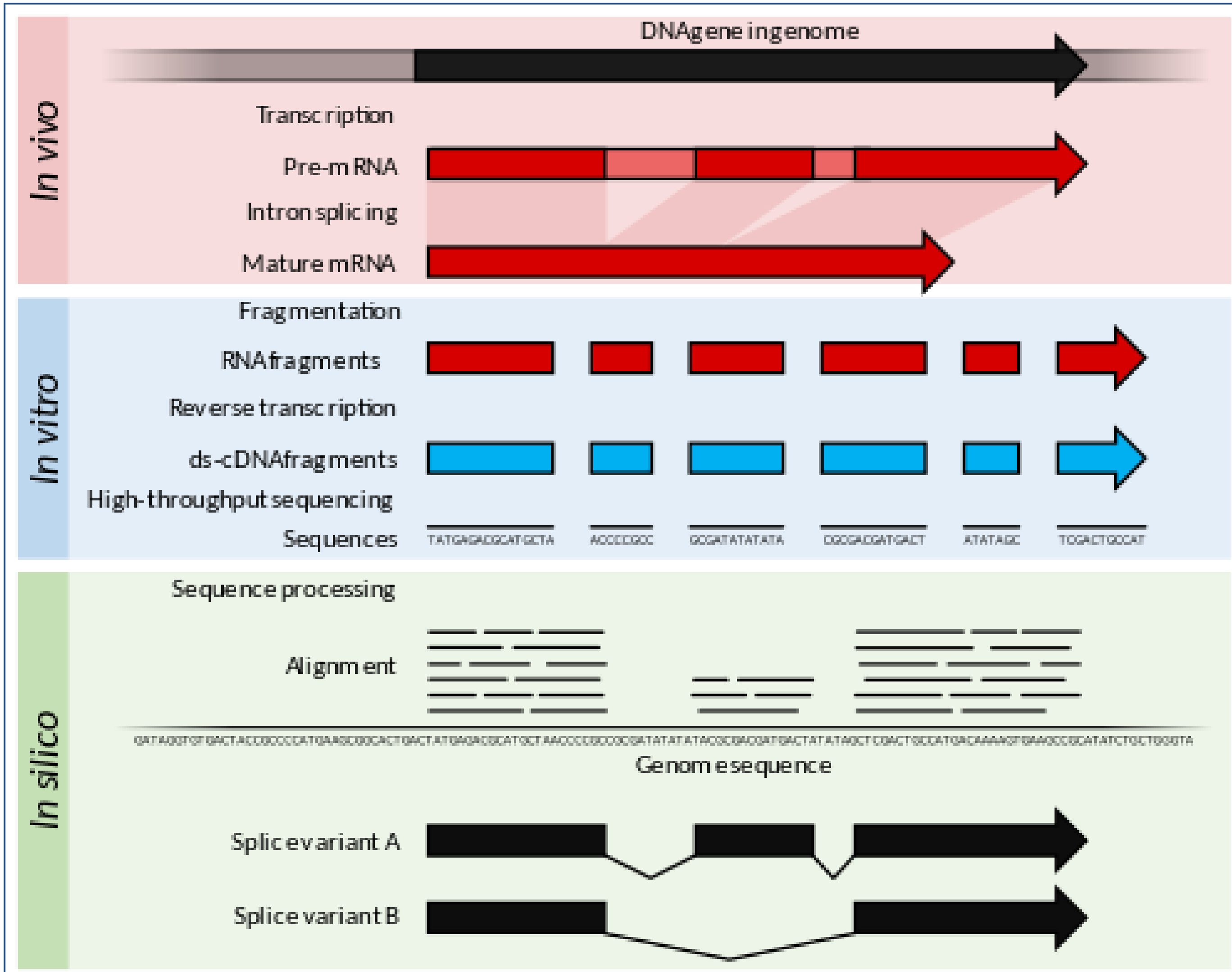
Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020 W. H. Freeman and Company

Look for cDNAs that map to the putative gene

Cells transcribe genes to make mRNA.

mRNA is isolated, reverse transcription generates cDNA copies of mRNAs, cDNAs are sequenced.

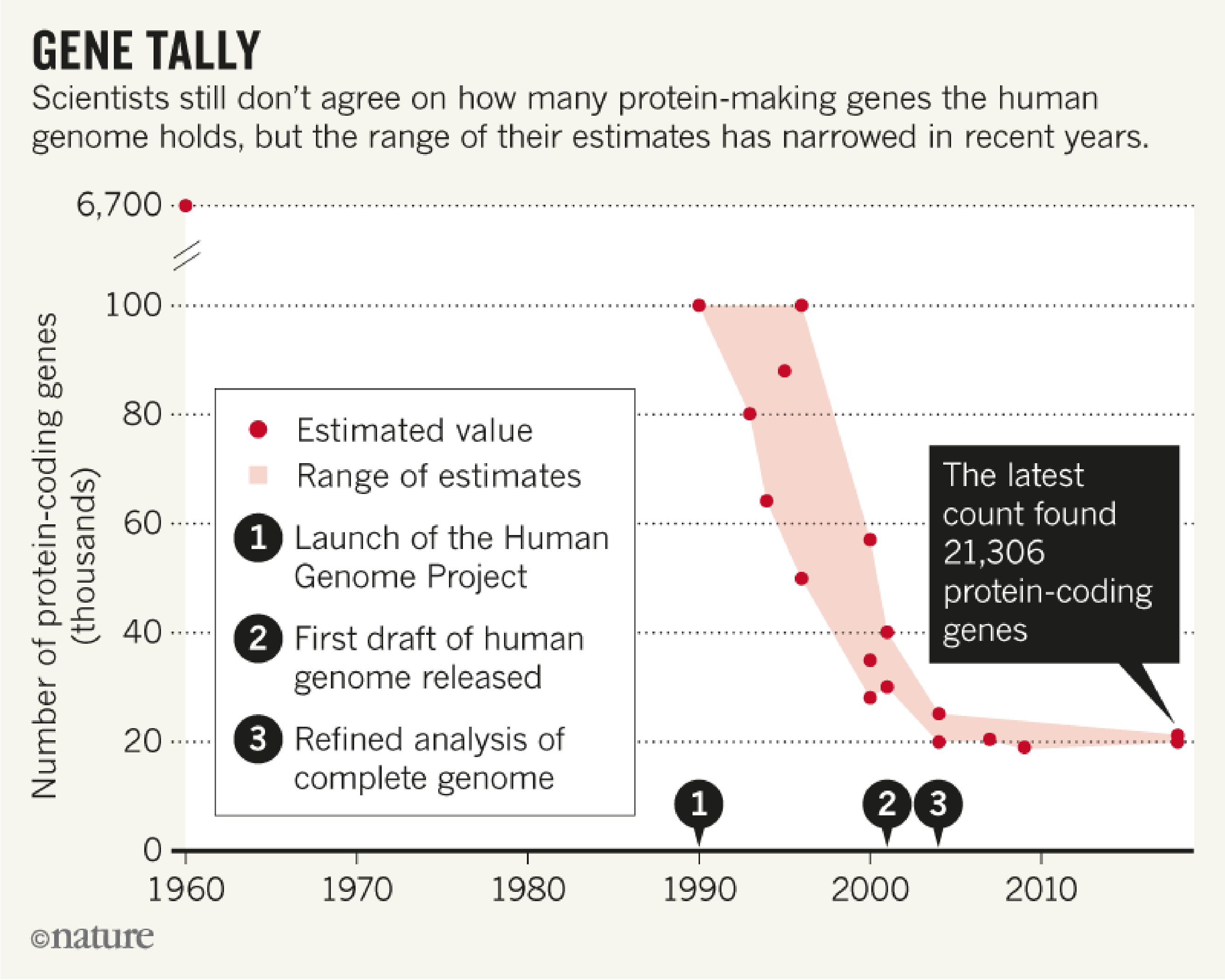
Sequences are aligned to the genome sequence to determine where the mRNA came from.



Think Break

The structure of the human genome

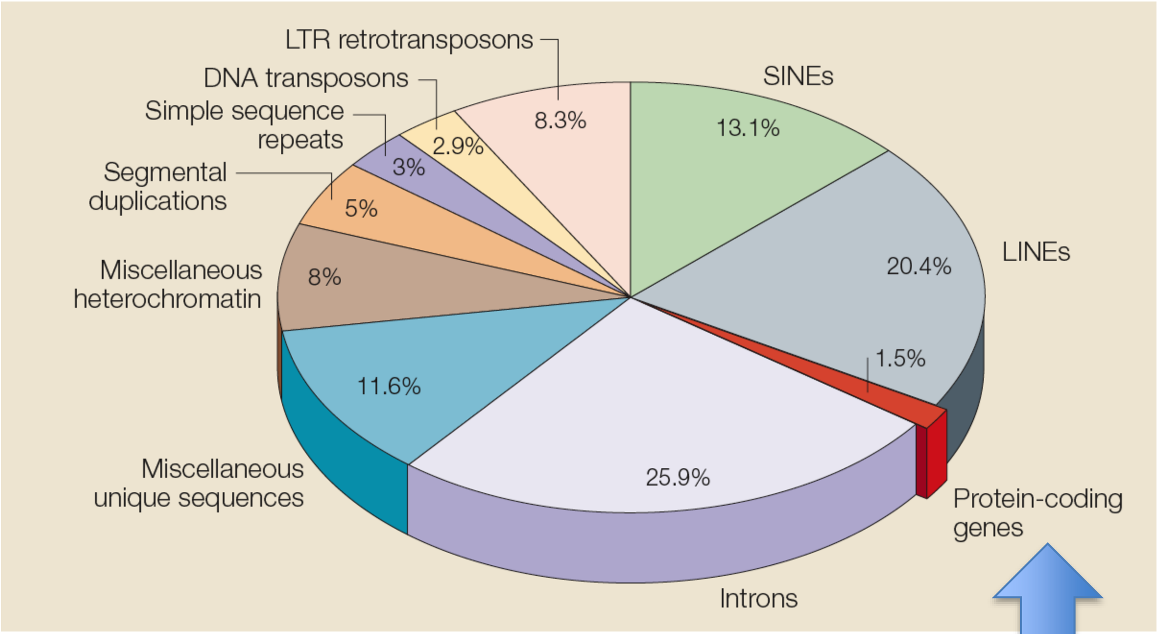
After all this analysis, how many genes are there?



New human gene tally reignites debate (nature.com) June 2018

The structure of the human genome

Protein-encoding genes are rare. Repeated sequences are common.

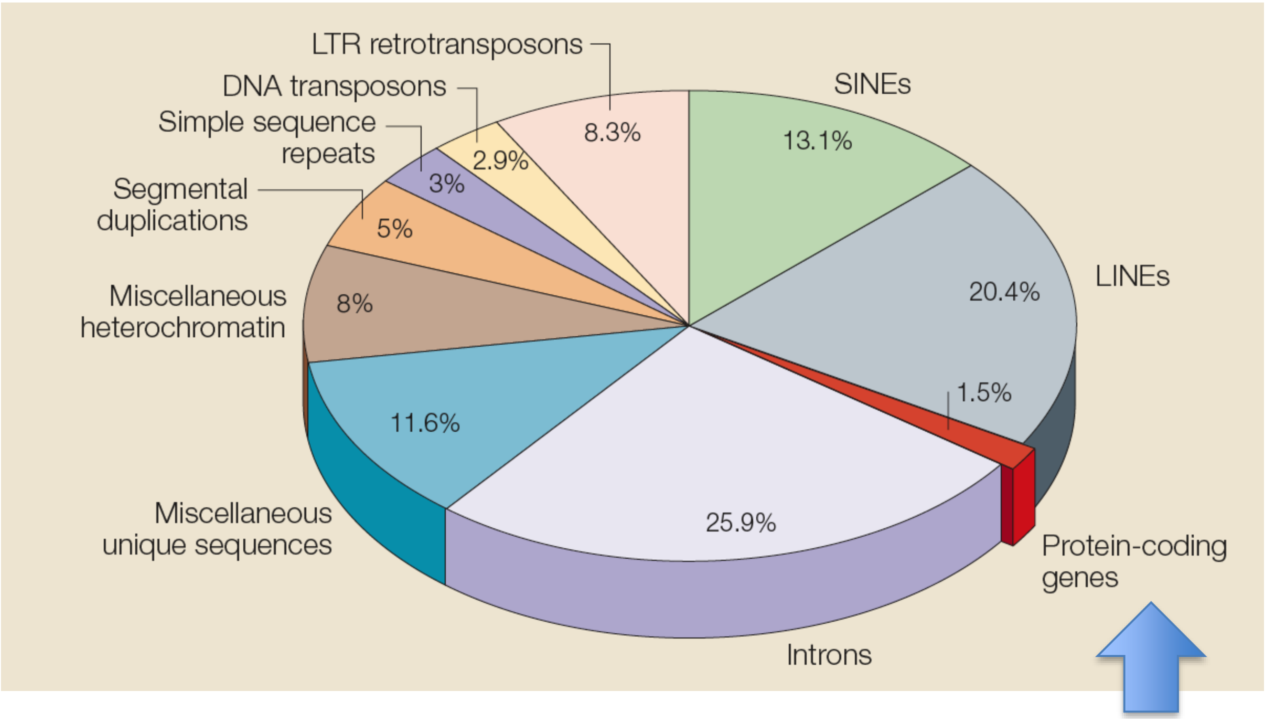


~3% of the human genome is made up of exons of protein-encoding genes.
(exons plus introns plus regulatory sequences: = ~28%)
~1% encodes protein sequences.
~45% is made up of repetitive sequences.

Think Break

The structure of the human genome

Protein-encoding genes are rare. Repeated sequences are common.



Beyond protein-encoding genes, what is the rest of the genome doing?
How can we figure this out?